

Surplus Food Redistribution Network

Frontend System Design & User Experience Documentation

BUILD Project

Mentor Me Collective × Grow with Google

UN Sustainable Development Goal 2: Zero Hunger

Team: Beacon Collaborative

CHAPTER ONE

Project Overview

1.1 Introduction

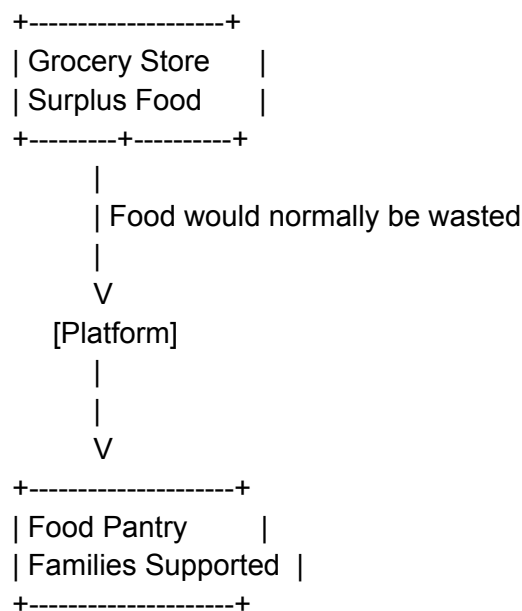
Food insecurity and food waste remain two significant challenges affecting communities worldwide. Every day, grocery stores, restaurants, bakeries, and other food providers dispose of large quantities of surplus food that is still safe for consumption, while many families and community organizations struggle to access nutritious meals.

According to the United Nations Sustainable Development Goal (SDG) 2: Zero Hunger, improving access to nutritious food while reducing unnecessary food waste is an essential step toward building healthier and more sustainable communities.

To address this challenge, Team Beacon Collaborative proposes the **Surplus Food Redistribution Network**, a digital platform designed to connect food donors with recipient organizations through an efficient, secure, and user-friendly web application.

The platform focuses on improving communication, simplifying food donation workflows, and reducing food waste by enabling organizations to redistribute surplus food before it expires.

Illustration 1.1 – Problem Overview



(In the final document, we'll replace this with a professional flow diagram.)

1.2 Background

Food waste has become an increasing concern in urban environments.

Many grocery stores regularly dispose of food that is close to its expiration date despite remaining safe for consumption.

At the same time, food pantries and community organizations frequently experience shortages that limit their ability to serve vulnerable populations.

Although numerous organizations work independently to reduce hunger, the lack of a centralized coordination platform often results in:

- delayed communication,
- missed donation opportunities,
- inefficient pickup scheduling,
- unnecessary food waste,
- unequal food distribution.

This project proposes a centralized digital solution that improves collaboration between donors and recipient organizations.

1.3 Problem Statement

Many grocery stores generate surplus food every day, yet there is no simple, centralized, and efficient platform that connects them with nearby food pantries in real time.

As a result,

- edible food is discarded,
- food pantries miss donation opportunities,
- coordination is inefficient,
- communities experiencing food insecurity receive less support than they could.

The challenge is particularly important within the six selected Chicago community areas identified during the team's research.



Illustration 1.2 – Current Problem

Store

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Food Surplus

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Thrown Away ❌

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Food Waste

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Families Still Hungry

1.4 Proposed Solution

The proposed solution is the **Surplus Food Redistribution Network**, a web-based platform that enables verified food donors and recipient organizations to coordinate food redistribution efficiently.

The platform introduces:

- centralized food listings,
- organization verification,
- role-based dashboards,
- interactive maps,
- pickup scheduling,
- notifications,
- reporting and analytics.

Rather than allowing food to become waste, the platform enables organizations to redistribute surplus food to communities that need it most.

The Minimum Viable Product (MVP) focuses on connecting **grocery stores** and **food pantries** within six selected Chicago community areas.

However, the system architecture is intentionally scalable, allowing future support for:

- restaurants,
- bakeries,
- farms,
- food manufacturers,
- soup kitchens,
- school meal programs,
- community organizations,
- regional food banks.

Illustration 1.3 – Proposed Solution

Donor

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Platform

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Recipient

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Food Pickup

↓

Community Served

1.5 Project Objectives

The primary objective of the project is to reduce food waste while improving food accessibility through technology.

Specific objectives include:

- Provide an easy-to-use platform for food donation.
 - Connect nearby donors and recipients.
 - Improve coordination through digital scheduling.
 - Reduce unnecessary food waste.
 - Support SDG Goal 2 (Zero Hunger).
 - Provide measurable impact through analytics.
 - Build a scalable platform capable of future expansion.
-

1.6 Project Scope

Current MVP

The first version of the platform focuses on:

Region

Chicago, Illinois, USA

Communities

The six community areas identified during the research.

Donors

- Grocery Stores

Recipients

- Food Pantries
-

Future Expansion

Future versions may support:

Donors

- Restaurants
- Bakeries
- Farms
- Food Manufacturers
- Supermarkets
- Community Organizations

Recipients

- Soup Kitchens
- School Meal Programs
- Community Organizations
- Regional Food Banks

The Information Architecture has already been designed to support this future growth without requiring major structural changes.

1.7 Alignment with SDG Goal 2

This project directly supports **United Nations Sustainable Development Goal 2: Zero Hunger** by improving food redistribution systems and reducing barriers between food donors and recipient organizations.

Rather than increasing food production, the platform focuses on maximizing the use of existing food resources that would otherwise be wasted.

Through technology, the platform contributes to:

- reducing hunger,
- minimizing food waste,
- strengthening local food networks,
- improving community resilience.

1.8 Expected Impact

If successfully implemented, the platform is expected to:

- Increase food redistribution.
- Reduce surplus food waste.
- Improve coordination between organizations.
- Increase the number of successful food pickups.
- Support vulnerable communities.
- Provide valuable data for measuring community impact.

Chapter Summary

This chapter introduced the motivation behind the Surplus Food Redistribution Network, explained the problem it addresses, outlined the proposed solution, defined the project's objectives and scope, and demonstrated its alignment with UN Sustainable Development Goal 2: Zero Hunger.

The following chapter will summarize the research findings that informed the design and development of the platform, providing the evidence base for the decisions documented throughout this report.

CHAPTER TWO

Research Summary

2.1 Introduction

The design and development of the **Surplus Food Redistribution Network** are grounded in research conducted to better understand food insecurity and food waste within selected Chicago communities. Rather than relying on assumptions, the project team analyzed available data to identify the communities most affected, understand existing challenges, and define a practical solution aligned with **United Nations Sustainable Development Goal (SDG) 2: Zero Hunger**.

The research provides the evidence that guides the platform's features, user roles, geographic scope, and overall system design.

Illustration 2.1 – Research-Driven Design

Research

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Problem Analysis

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Key Findings

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Platform Requirements

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System Design

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Frontend Development

2.2 Research Objectives

The research was conducted to answer the following questions:

- Which communities experience the greatest need for food assistance?
- What challenges prevent efficient food redistribution?
- Which organizations are involved in food donation and food distribution?
- What barriers exist between food donors and recipients?
- How can technology improve food redistribution?

The answers to these questions formed the basis for the platform's requirements.

2.3 Geographic Scope

The project focuses on **Chicago, Illinois**, specifically six community areas identified in the research as priority locations.

The decision to limit the Minimum Viable Product (MVP) to these communities allows the team to develop a realistic, manageable, and evidence-based solution rather than attempting to solve food insecurity across an entire city or country in the first version.

The architecture, however, has been intentionally designed to support future expansion into additional communities, cities, and regions.

Illustration 2.2 – MVP Scope

United States

↓

Illinois

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Chicago

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Six Priority Community Areas

↓

MVP Platform

2.4 Key Problems Identified

The research identified several recurring challenges affecting food redistribution.

1. Food Waste

Large quantities of edible food are discarded before they can be redistributed to people in need.

Examples include:

- Fresh produce
- Bread
- Dairy products
- Packaged foods
- Prepared meals that remain safe for consumption

This represents both an environmental challenge and a missed opportunity to reduce hunger.

2. Food Insecurity

Many families continue to experience difficulty accessing nutritious food despite the availability of surplus food from businesses.

Food pantries and other recipient organizations often struggle to meet increasing demand with limited resources.

3. Lack of Coordination

One of the most significant findings is the absence of a centralized coordination system.

Without an efficient platform:

- donors may not know where to donate,
 - recipient organizations may not know what food is available,
 - pickups may be delayed,
 - food may expire before collection.
-

4. Limited Communication

Communication between food donors and recipient organizations often relies on phone calls, emails, or informal arrangements.

This creates delays and increases the likelihood that food will be wasted.

Illustration 2.3 – Current Challenges

Food Surplus

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Poor Communication

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Delayed Pickup

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Food Expires

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Food Waste

2.5 Opportunities Identified

While the research highlighted significant challenges, it also revealed opportunities for improvement.

These include:

- Better communication between organizations.
- Faster identification of available donations.
- Improved pickup coordination.
- Real-time visibility of available food.
- Digital tracking of food redistribution.
- Data-driven monitoring of community impact.

These opportunities informed the proposed solution.

2.6 Research Findings and Design Decisions

Every major feature of the platform can be traced directly to a research finding.

Research Finding	Design Decision
Lack of coordination	Centralized food listing platform
Communication delays	Built-in messaging and notifications
Food expires before pickup	Pickup scheduling
Difficulty locating nearby donations	Interactive map
Need to measure impact	Analytics dashboard
Different organization responsibilities	Role-based dashboards

This traceability ensures that the platform solves real problems rather than adding unnecessary features.

2.7 Stakeholders Identified

The research indicates that several groups play important roles in food redistribution.

Primary Stakeholders

- Grocery Stores (Donors)

- Food Pantries (Recipients)

These organizations form the core of the MVP.

Secondary Stakeholders

Future versions of the platform may include:

Donors

- Restaurants
- Bakeries
- Farms
- Food Manufacturers
- Supermarkets
- Community Organizations

Recipients

- Soup Kitchens
- School Meal Programs
- Community Organizations
- Regional Food Banks

Support Roles

- Volunteers
- Platform Administrators

This stakeholder analysis influenced the role-based architecture of the platform.

2.8 Success Metrics

The research emphasizes the importance of measuring impact.

The platform should therefore support key performance indicators (KPIs) such as:

- Number of registered donor organizations
- Number of registered recipient organizations
- Number of food donations created
- Number of successful pickups

- Quantity of food redistributed
- Estimated reduction in food waste
- Communities served

These metrics will be displayed within the administrator dashboard and selected public impact pages.



Illustration 2.4 – Measuring Success

Food Donations



Successful Pickups



Food Redistributed



Reduced Food Waste



Improved Community Food Access

2.9 How the Research Influenced the Frontend

The frontend design decisions are directly informed by the research.

Examples include:

- **Role-based dashboards** ensure donors and recipients only see features relevant to their responsibilities.
- **Interactive maps** improve awareness of participating organizations while protecting sensitive donation details.
- **Notification systems** address communication delays.

- **Pickup scheduling** helps reduce food spoilage caused by late collections.
- **Public impact pages** communicate the platform's value and encourage additional organizations to participate.

By grounding the interface in research findings, the platform becomes more usable and purpose-driven.

2.10 Research Limitations

As with any MVP, the project has defined boundaries.

The current implementation is limited to:

- Chicago, Illinois.
- The six community areas identified in the research.
- Grocery stores as donor organizations.
- Food pantries as recipient organizations.

These limitations are intentional and help ensure the project remains achievable while leaving room for future expansion.

Chapter Summary

The research established a clear understanding of the food redistribution challenges faced within the selected Chicago communities. It identified food waste, communication gaps, and coordination inefficiencies as key problems and highlighted opportunities for improvement through technology.

These findings directly influenced the platform's architecture, feature set, navigation, and user experience. Every major component of the Surplus Food Redistribution Network can be traced back to evidence gathered during the research phase, ensuring that the proposed solution remains practical, scalable, and aligned with the project's objectives.

CHAPTER THREE

Product Vision

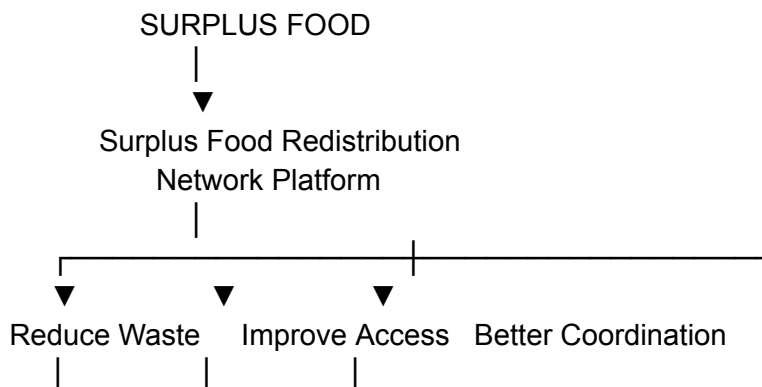
3.1 Introduction

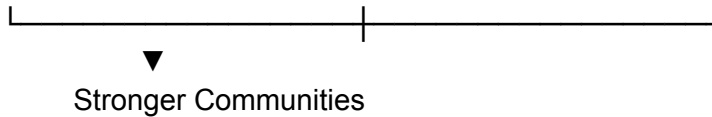
The **Surplus Food Redistribution Network** is envisioned as a digital platform that simplifies the process of redistributing surplus food from donor organizations to recipient organizations. The platform aims to reduce food waste, improve food accessibility, and strengthen collaboration between organizations working to address food insecurity.

Rather than serving as a simple donation listing website, the platform functions as a centralized coordination system that enables verified organizations to communicate, schedule pickups, monitor donations, and measure community impact.

The design of the platform is informed by the research findings and aligns with **United Nations Sustainable Development Goal 2 (Zero Hunger)** by using technology to improve the efficiency of food redistribution.

Illustration 3.1 – Product Vision





3.2 Vision Statement

Vision

To create a secure, scalable, and user-friendly digital platform that connects food donors with recipient organizations, making surplus food redistribution faster, more efficient, and more accessible while reducing food waste and supporting communities experiencing food insecurity.

This vision reflects both the immediate objectives of the project and its long-term potential for growth beyond the initial pilot implementation.

3.3 Mission Statement

Mission

To provide an easy-to-use digital platform that enables verified organizations to coordinate food donations efficiently through centralized food listings, real-time communication, location-aware services, and role-based management tools.

The mission emphasizes practical implementation rather than theoretical solutions, ensuring that organizations can quickly connect and redistribute food before it becomes waste.

3.4 Product Goals

The platform has been designed with several primary goals.

Goal 1 – Reduce Food Waste

A significant amount of edible food is discarded each day due to inefficient redistribution processes.

The platform seeks to reduce unnecessary waste by allowing donors to publish available food quickly and efficiently.

Goal 2 – Improve Food Accessibility

Many communities experience food insecurity despite the availability of surplus food.

The platform helps recipient organizations identify and reserve available donations before they expire.

Goal 3 – Improve Coordination

The research identified communication delays as one of the largest challenges.

The platform therefore introduces:

- centralized food listings,
- notifications,
- messaging,
- pickup scheduling,
- organization verification.

Together, these features reduce coordination delays.

Goal 4 – Support Data-Driven Decisions

The platform provides administrators with analytics and reporting tools to monitor community impact, evaluate platform performance, and measure key performance indicators (KPIs).

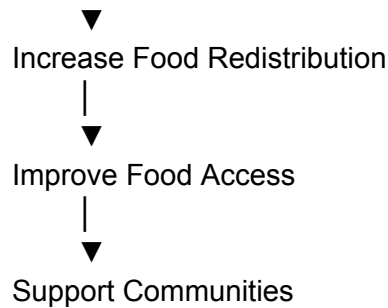
Illustration 3.2 – Product Goals

Reduce Food Waste



Improve Coordination





3.5 Why a Digital Platform?

Traditional food redistribution often depends on manual communication methods such as phone calls, emails, or personal relationships between organizations.

While these approaches may work on a small scale, they become increasingly inefficient as the number of participating organizations grows.

A centralized digital platform provides several advantages:

- Real-time food availability.
- Faster communication.
- Easier coordination.
- Transparent donation tracking.
- Better reporting.
- Improved scalability.

By digitizing these processes, organizations can spend less time coordinating logistics and more time serving their communities.

3.6 Minimum Viable Product (MVP)

The project follows a **Minimum Viable Product (MVP)** approach.

Rather than attempting to solve every aspect of food redistribution from the beginning, the first version focuses on delivering a practical solution that addresses the most important problems identified in the research.

Geographic Scope

- Chicago, Illinois

Communities

- Six selected community areas identified during research

Donor Organizations

- Grocery Stores

Recipient Organizations

- Food Pantries

This limited scope enables the team to build, test, and evaluate a working solution before expanding to additional organization types or geographic regions.

Illustration 3.3 – MVP Scope



3.7 Future Vision

Although the MVP focuses on grocery stores and food pantries, the platform has been intentionally designed to support future expansion.

Future Donor Organizations

- Restaurants
- Bakeries
- Farms
- Food Manufacturers
- Supermarkets
- Community Organizations

Future Recipient Organizations

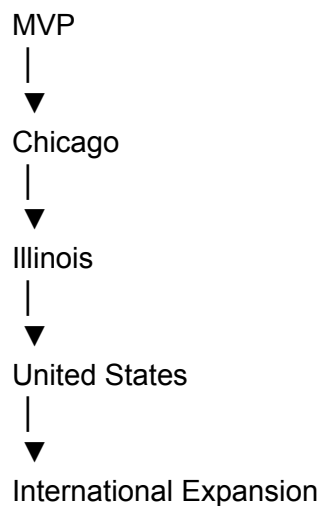
- Soup Kitchens
- School Meal Programs
- Community Organizations
- Regional Food Banks

Additional Future Features

- Volunteer delivery coordination.
- AI-assisted donation recommendations.
- Predictive food expiration alerts.
- Multi-city deployment.
- Advanced analytics and reporting.
- Mobile application support.

The Information Architecture and system design have been developed to accommodate these future enhancements without requiring significant structural changes.

Illustration 3.4 – Product Growth



3.8 Guiding Design Principles

The frontend design of the platform follows several guiding principles.

Simplicity

Users should be able to complete common tasks with minimal effort.

Accessibility

The platform should remain usable by individuals with varying levels of technical experience and should follow accessibility best practices wherever possible.

Scalability

The platform architecture should support future expansion without requiring major redesigns.

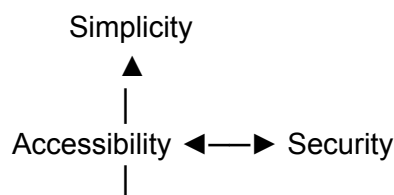
Security

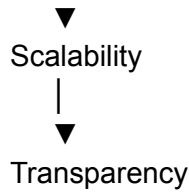
Only verified organizations should access sensitive operational features such as food reservations, donor information, and pickup scheduling.

Transparency

The platform should clearly communicate its impact through statistics, reports, and public-facing information while protecting sensitive operational data.

Illustration 3.5 – Design Principles





3.9 Success Vision

The long-term success of the Surplus Food Redistribution Network will not be measured solely by the number of registered users but by its measurable impact on food redistribution and community support.

Indicators of success include:

- Increased food donations.
- Reduced food waste.
- Improved coordination between organizations.
- Faster pickup scheduling.
- Greater community participation.
- Expansion to additional donor and recipient organization types.
- Replication of the platform in additional cities and regions.

These outcomes align with both the project's objectives and the broader goals of sustainable food management.

Chapter Summary

This chapter established the vision, mission, and long-term direction of the Surplus Food Redistribution Network. It explained why a centralized digital platform is needed, defined the goals of the Minimum Viable Product (MVP), and outlined how the system has been designed for future scalability. The guiding design principles introduced in this chapter will inform all subsequent decisions related to user experience, interface design, navigation, and system architecture.

CHAPTER FOUR

System Overview

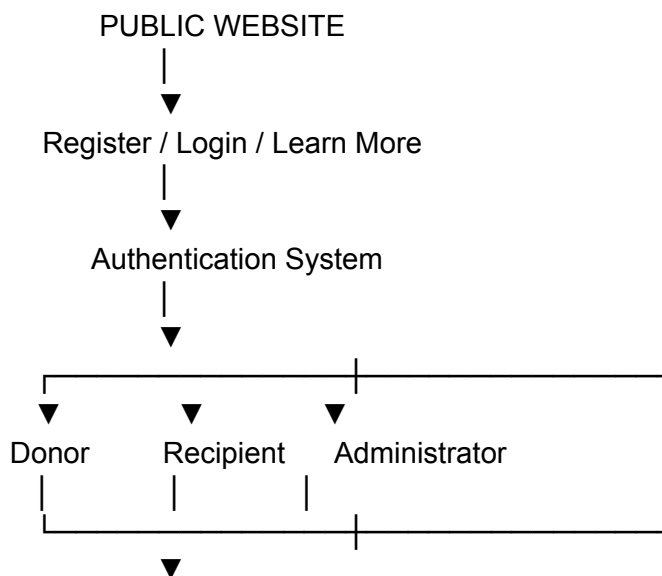
4.1 Introduction

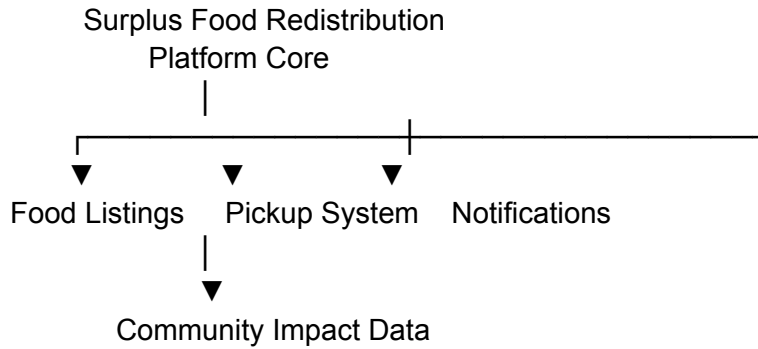
The **Surplus Food Redistribution Network** is a web-based platform designed to facilitate the redistribution of surplus food between verified donor and recipient organizations. The platform serves as a centralized coordination hub where organizations can create food donations, discover available food, schedule pickups, communicate with one another, and monitor community impact.

The system has been designed with scalability, usability, and security in mind. While the initial deployment focuses on six selected community areas within Chicago, Illinois, the overall architecture supports future expansion to additional communities, cities, and organization types.

Unlike traditional food donation processes that rely on manual communication, the platform digitizes the complete workflow, enabling faster coordination and reducing food waste.

Illustration 4.1 – High-Level System Overview





4.2 System Purpose

The primary purpose of the platform is to provide an efficient digital environment where organizations can collaborate to reduce food waste and improve food accessibility.

The system enables:

- Food donors to publish surplus food.
- Recipient organizations to discover and reserve available food.
- Volunteers (future implementation) to assist with transportation.
- Administrators to monitor platform activity and ensure proper operation.

The platform acts as the central point of communication between all participating organizations.

4.3 System Users

The platform supports four primary user roles.

Donor

Current MVP:

- Grocery Store

Future Expansion:

- Restaurant

- Bakery
- Farm
- Food Manufacturer
- Supermarket
- Community Organization

Primary Responsibilities:

- Publish food donations.
 - Manage donations.
 - Schedule pickups.
 - Monitor donation history.
-

Recipient

Current MVP:

- Food Pantry

Future Expansion:

- Soup Kitchen
- School Meal Program
- Community Organization
- Regional Food Bank

Primary Responsibilities:

- Browse available food.
 - Reserve donations.
 - Schedule pickups.
 - Track reservation history.
-

Volunteer (Future)

Primary Responsibilities:

- Accept delivery assignments.
- Pick up food.
- Deliver food.
- Update delivery status.

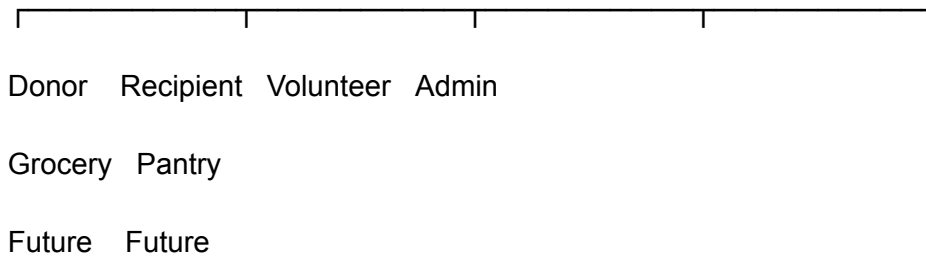
Administrator

Responsibilities:

- Manage users.
 - Verify organizations.
 - Monitor donations.
 - View analytics.
 - Manage community coverage.
 - Maintain platform integrity.
-

Illustration 4.2 – User Roles

USERS



4.4 System Modules

The platform is organized into several functional modules.

Public Website Module

Purpose:

Provide general information about the platform and encourage organizations to register.

Includes:

- Home
 - About
 - How It Works
 - Impact
 - Contact
 - Login
 - Register
-

Authentication Module

Responsible for:

- Registration
- Login
- Role assignment
- Account verification
- Password management

This module ensures only verified users can access protected platform features.

Donor Module

Allows donor organizations to:

- Create donations.
 - Edit donations.
 - Upload food images.
 - Manage food listings.
 - Track pickup schedules.
 - Receive notifications.
-

Recipient Module

Allows recipient organizations to:

- Search available food.
- Reserve donations.
- Schedule pickups.

- Receive notifications.
 - Track reservation history.
-

Administration Module

Provides tools for administrators to:

- Manage users.
 - Verify organizations.
 - Monitor platform activity.
 - Generate reports.
 - View analytics.
 - Manage supported community areas.
-

Notification Module

Handles communication between organizations.

Examples include:

- New donation available.
 - Reservation confirmed.
 - Pickup reminder.
 - System announcements.
-

Mapping Module

Displays participating organizations geographically.

Public users can view:

- Supported communities.
- Registered organizations.

Verified users can additionally view:

- Available donations.
- Pickup locations.

- Organization details appropriate to their role.
-

Analytics Module

Collects operational statistics including:

- Food redistributed.
 - Active donations.
 - Completed pickups.
 - Registered organizations.
 - Community impact.
-

4.5 Overall System Workflow

The system follows a straightforward operational process.

Step 1

A donor organization logs into the platform.

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Step 2

The donor creates a food donation.

↓

Step 3

The donation becomes available.

↓

Step 4

Nearby recipient organizations receive notifications.

↓

Step 5

A recipient reserves the donation.

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Step 6

Pickup is scheduled.

↓

Step 7

Food is collected.

↓

Step 8

Donation is completed.

↓

Step 9

Platform statistics are updated.



Illustration 4.3 – Overall Workflow

Donor

↓

Create Donation

↓

Platform

↓

Recipients Notified

↓

Reserve Food

↓

Pickup Scheduled

↓

Food Delivered

↓

Impact Updated

4.6 Public and Private Platform

The platform is divided into two distinct environments.

Public Platform

Accessible without authentication.

Purpose:

- Introduce the project.
- Explain the mission.
- Display public statistics.
- Show participating communities.
- Encourage registration.

Public users cannot:

- View active donations.
 - Reserve food.
 - Schedule pickups.
 - Access dashboards.
-

Private Platform

Accessible only after authentication.

Users receive dashboards tailored to their assigned role.

Private users can perform operational tasks such as:

- Creating donations.
- Reserving food.
- Messaging organizations.
- Managing pickups.
- Viewing notifications.

This separation improves both usability and security.

4.7 Scalability

Although the MVP focuses on grocery stores and food pantries, the system has been designed to support future expansion.

Future enhancements may include:

- Additional donor categories.
- Additional recipient categories.
- Volunteer coordination.
- Multi-city support.
- Mobile applications.
- AI-assisted recommendations.
- Advanced reporting.

The modular architecture allows these features to be introduced with minimal impact on existing functionality.

Illustration 4.4 – System Growth

MVP

↓

More Donors



More Recipients



More Communities



More Cities



National Platform

4.8 Security Considerations

Security is fundamental to the platform.

Key design decisions include:

- Role-based access control.
- Secure authentication.
- Organization verification.
- Protected operational data.
- Public/private content separation.
- Administrative oversight.

Sensitive information such as active food listings, reservations, and user management functions is available only to authenticated and authorized users.

4.9 Relationship Between Research and System Design

The overall system architecture directly reflects the findings of the research.

Research Finding	System Response
Food waste	Donation management module
Communication delays	Notification and messaging modules
Limited coordination	Centralized platform
Geographic challenges	Community map integration
Need for accountability	Analytics and reporting
Different organization responsibilities	Role-based dashboards

This traceability ensures that every major system component addresses a documented need.

Chapter Summary

This chapter introduced the overall architecture of the Surplus Food Redistribution Network. It described the platform's purpose, user roles, functional modules, operational workflow, public and private environments, scalability strategy, and security considerations. By establishing the system's high-level structure, this chapter provides the foundation for the detailed frontend design decisions presented in the following chapters.

CHAPTER FIVE

User Personas

5.1 Introduction

Designing an effective digital platform begins with understanding the people who will use it. The Surplus Food Redistribution Network serves multiple stakeholders with different responsibilities, goals, and challenges. To ensure the platform meets their needs, user personas were developed based on the project research and intended system functionality.

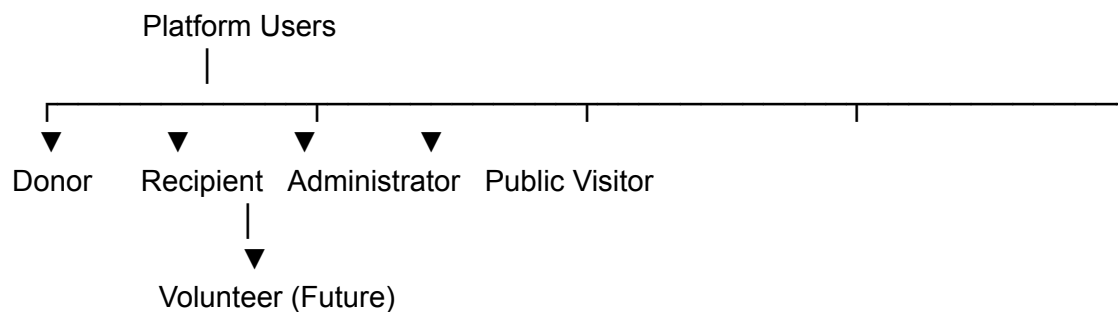
A **user persona** is a fictional representation of a typical user group. Personas help guide design decisions by keeping the focus on user needs rather than technical implementation.

For the Minimum Viable Product (MVP), four primary personas have been identified:

- Donor
- Recipient
- Administrator
- Public Visitor

A future version of the platform will also include a Volunteer persona.

Illustration 5.1 – Platform Users



5.2 Persona 1 – Donor

Overview

The donor represents organizations that have surplus food available for redistribution. During the MVP, donors consist primarily of grocery stores operating within the selected Chicago communities.

Future versions of the platform will support restaurants, bakeries, farms, supermarkets, food manufacturers, and other food providers.

Persona Profile

Attribute	Description
Persona Name	Grocery Store Manager
Role	Donor
Organization	Grocery Store
Technical Experience	Moderate
Primary Device	Desktop / Tablet
Goals	Reduce food waste, donate efficiently
Main Tasks	Create donations, manage listings, schedule pickups

Goals

The donor wants to:

- Reduce edible food waste.
 - Donate food quickly before expiration.
 - Coordinate pickups efficiently.
 - Track previous donations.
 - View the impact of donations.
-

Pain Points

The donor currently experiences:

- Difficulty identifying recipient organizations.
 - Time-consuming communication.
 - Missed pickup opportunities.
 - Food expiring before collection.
 - Limited visibility into donation impact.
-

Needs

The platform should allow donors to:

- Create donations quickly.
 - Upload food details.
 - Receive reservation notifications.
 - Manage donation history.
 - Schedule pickups.
 - View impact statistics.
-

User Story

As a grocery store manager, I want to list surplus food so that nearby food pantries can collect it before it becomes waste.

Illustration 5.2 – Donor Journey

Surplus Food

↓

Create Donation

↓

Recipients Notified

↓

Food Reserved



Pickup Scheduled



Donation Completed

5.3 Persona 2 – Recipient

Overview

Recipients are organizations responsible for collecting and distributing donated food to individuals and families experiencing food insecurity.

For the MVP, recipients are food pantries operating within the selected Chicago communities.

Future versions of the platform may include soup kitchens, school meal programs, community organizations, and regional food banks.

Persona Profile

Attribute	Description
Persona Name	Food Pantry Coordinator
Role	Recipient
Organization	Food Pantry
Technical Experience	Moderate
Primary Device	Desktop / Mobile
Goals	Locate available food and coordinate pickups

Goals

The recipient wants to:

- Find nearby food donations.
 - Reserve available food.
 - Coordinate pickups.
 - Monitor reservation history.
 - Reduce response time.
-

Pain Points

Current challenges include:

- Limited awareness of available donations.
 - Communication delays.
 - Manual coordination.
 - Competition for available food.
 - Transportation scheduling.
-

Needs

The platform should provide:

- Searchable food listings.
 - Interactive map.
 - Reservation system.
 - Pickup scheduling.
 - Notification system.
-

User Story

As a food pantry coordinator, I want to receive notifications about available food so that I can reserve donations before they expire.



Illustration 5.3 – Recipient Journey

Notification



Browse Food



Reserve Donation



Schedule Pickup



Collect Food



Serve Community

5.4 Persona 3 – Administrator

Overview

Administrators are responsible for ensuring the smooth operation of the platform. They oversee organization verification, user management, analytics, and system monitoring.

Unlike donors and recipients, administrators do not directly participate in food redistribution. Instead, they maintain the integrity and effectiveness of the platform.

Persona Profile

Attribute	Description
Persona Name	Platform Administrator
Role	Administrator

Technical Experience	High
Primary Device	Desktop
Goals	Maintain platform reliability and security

Goals

The administrator wants to:

- Verify organizations.
 - Manage user accounts.
 - Monitor platform activity.
 - Review analytics.
 - Resolve issues.
 - Ensure data integrity.
-

Pain Points

Administrators must manage:

- Fake registrations.
 - Incorrect listings.
 - User disputes.
 - Platform monitoring.
 - System performance.
-

Needs

The platform should include:

- Verification tools.
 - User management dashboard.
 - Analytics dashboard.
 - Reporting system.
 - Moderation tools.
-

User Story

As a platform administrator, I want to verify organizations before granting access so that only trusted users can participate in food redistribution.

Illustration 5.4 – Administrator Workflow

User Registers

↓

Verification

↓

Approve / Reject

↓

Platform Access

↓

Monitor Activity

↓

Generate Reports

5.5 Persona 4 – Public Visitor

Overview

Not every visitor to the website will immediately register. Many users will first want to understand the purpose of the platform, explore its impact, and determine whether it is relevant to their organization.

The public website serves as an introduction to the platform while encouraging eligible organizations to join.

Persona Profile

Attribute	Description
Persona Name	Community Visitor
Role	Public User
Goals	Learn about the platform

Goals

A public visitor wants to:

- Understand the platform.
 - Learn how food redistribution works.
 - View community impact.
 - Discover participating organizations.
 - Decide whether to register.
-

Needs

The public website should provide:

- Clear project overview.
 - Interactive community map (without sensitive donation details).
 - Impact statistics.
 - Frequently Asked Questions.
 - Registration information.
-

User Story

As a community visitor, I want to understand how the platform works before deciding whether my organization should participate.

Illustration 5.5 – Public Visitor Journey

Visit Website

↓

Learn About Project

↓

View Impact

↓

Explore Communities

↓

Register Organization

5.6 Future Persona – Volunteer

Although volunteers are not included in the MVP, the platform architecture has been designed to support them in future releases.

Volunteers would assist with transportation and delivery logistics when donor and recipient organizations require additional support.

Potential volunteer capabilities include:

- Accept delivery assignments.
- View pickup routes.
- Update delivery status.
- Communicate with participating organizations.

Designing for future personas at this stage ensures the platform remains scalable.

5.7 Persona Comparison

Persona	Primary Goal	Main Dashboard	Key Features
---------	--------------	----------------	--------------

Donor	Donate surplus food	Donor Dashboard	Create donations, manage listings, pickup scheduling
Recipient	Receive available food	Recipient Dashboard	Browse donations, reserve food, pickup scheduling
Administrator	Manage platform	Admin Dashboard	Verification, analytics, user management
Public Visitor	Learn about the platform	Public Website	Information, impact, registration

5.8 How Personas Influenced the Design

The personas directly shaped the frontend design decisions.

For example:

- **Donors** need a fast donation form, so the interface prioritizes quick data entry.
- **Recipients** need rapid access to available food, leading to searchable listings and map integration.
- **Administrators** require dashboards with analytics, verification tools, and moderation features.
- **Public visitors** need informative content, resulting in a marketing-style homepage with impact statistics and clear calls to action.

These personas ensure that every screen and feature supports a real user need identified during the research.

Chapter Summary

This chapter defined the primary users of the Surplus Food Redistribution Network and documented their goals, responsibilities, pain points, and interactions with the platform. These personas provide the foundation for all subsequent design decisions, ensuring the interface remains user-centered and aligned with the project's objectives.

CHAPTER SIX

Information Architecture

6.1 Introduction

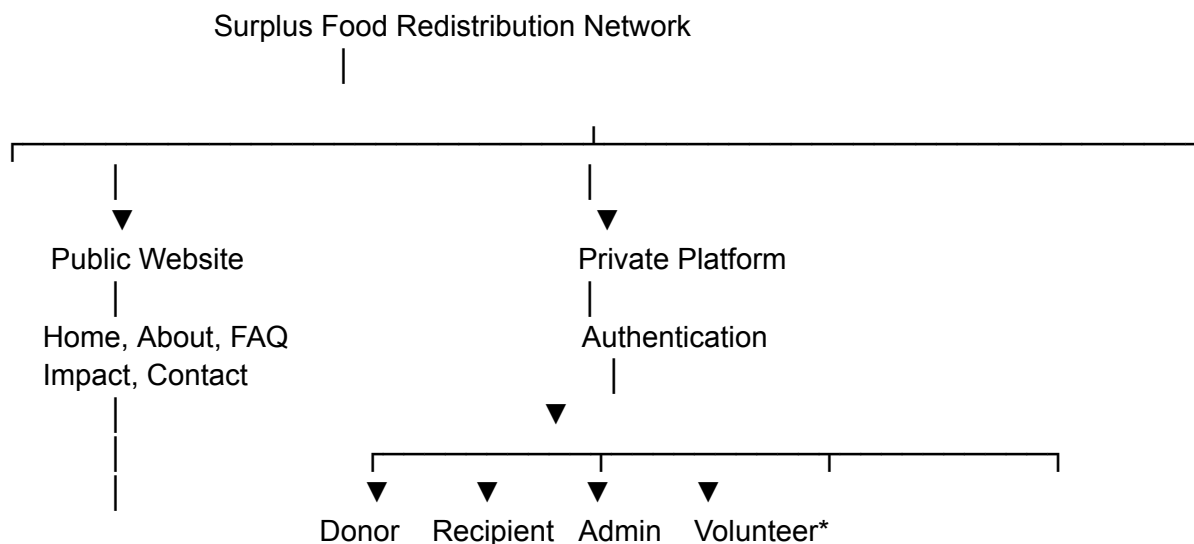
Information Architecture (IA) defines how information, pages, features, and user interactions are organized within the Surplus Food Redistribution Network. A well-designed information architecture enables users to navigate the platform efficiently, complete tasks with minimal effort, and quickly locate the information they need.

The architecture of this platform has been designed around the project's research findings, user personas, and product vision. It separates the platform into two primary environments:

- **Public Website** – Accessible without authentication.
- **Private Platform** – Accessible only to verified users through role-based dashboards.

This separation ensures that public visitors can learn about the platform while operational activities remain secure and available only to authorized organizations.

Illustration 6.1 – Overall Information Architecture



Volunteer dashboard reserved for future expansion.

6.2 Architecture Principles

The platform architecture follows several principles.

User-Centered

Every page is designed around a specific user goal rather than around technical features.

Simplicity

Users should complete common tasks in as few steps as possible.

Scalability

The architecture supports additional cities, organizations, and user roles without restructuring the application.

Security

Sensitive operational pages are separated from public content using authentication and role-based access control.

Consistency

Navigation, layout, colors, typography, and interaction patterns remain consistent throughout the platform.

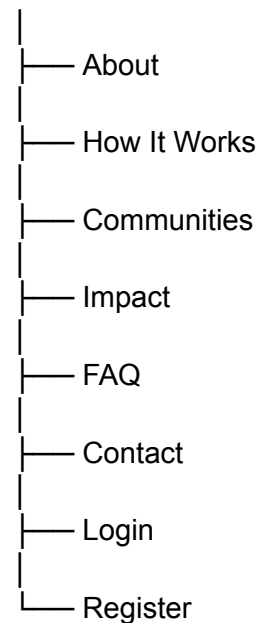
6.3 Public Website Architecture

The public website serves as the entry point for visitors. Its purpose is to educate, build trust, and encourage eligible organizations to register.

Unlike the dashboard, the public website does **not** expose operational information such as active donations or reservation details.

Public Website Structure

Home



Home

Purpose:

- Introduce the platform.
- Explain the mission.
- Encourage registration.
- Highlight community impact.

About

Purpose:

Explain:

- Why the platform exists.
 - SDG Goal 2.
 - Food waste problem.
 - Team vision.
-

How It Works

Purpose:

Explain the complete donation process from donor to recipient.

Communities

Purpose:

Display:

- Six Chicago community areas.
- Participating organizations.
- Public map.

No donation information is displayed here.

Impact

Purpose:

Display public statistics such as:

- Food redistributed.
 - Organizations participating.
 - Donations completed.
 - Communities served.
-

FAQ

Purpose:

Answer common questions.

Examples:

- Who can register?
 - How are organizations verified?
 - Is the platform free?
-

Contact

Purpose:

Allow visitors to communicate with administrators.

Login

Purpose:

Authenticate registered users.

Register

Purpose:

Allow organizations to create accounts.



Illustration 6.2 – Public Website Sitemap

Home

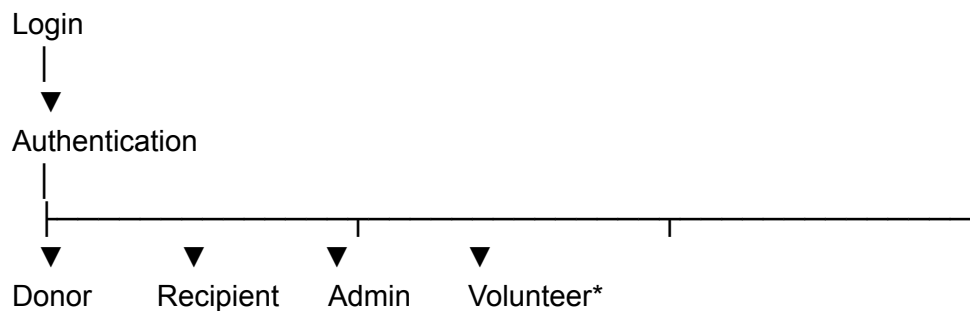


6.4 Private Platform Architecture

Once authenticated, users are redirected to dashboards based on their assigned role.

Each role has a unique navigation structure containing only the features relevant to that user.

Illustration 6.3 – Role-Based Access



6.5 Donor Information Architecture

The donor dashboard is designed to help organizations publish and manage surplus food quickly.



Dashboard Overview

Displays:

- Donation summary.
 - Upcoming pickups.
 - Recent activity.
 - Quick actions.
-

Donate Food

Allows donors to:

- Upload food.
 - Add quantity.
 - Add expiry date.
 - Add pickup window.
 - Upload images.
-

My Donations

Displays:

- Active donations.
 - Completed donations.
 - Cancelled donations.
 - Donation history.
-

Pickup Schedule

Displays upcoming collection schedules.

Notifications

Displays:

- Reservation confirmations.
- Pickup reminders.
- System updates.

Messages

Communication between donors and recipients.

Profile

Manage organization information.

Settings

Account preferences and security settings.

6.6 Recipient Information Architecture

Recipients have a different workflow.

Dashboard

- |
 - Overview
 - Available Food
 - My Reservations
 - Pickup Schedule
 - Notifications
 - Messages
 - Profile
 - Settings
 - Logout
-

Available Food

Search.

Filter.

Reserve.

View food details.

My Reservations

Displays:

- Pending
- Approved
- Completed

Pickup Schedule

Displays confirmed pickup appointments.

Notifications

Reservation updates.

Donation alerts.

Messages

Communication with donors.

6.7 Administrator Information Architecture

Administrators require more management tools.

Dashboard

- |
- Overview
- Users
- Organizations
- Food Listings

- Communities
 - Reports
 - Analytics
 - Notifications
 - Settings
 - Logout
-

Users

Manage user accounts.

Organizations

Verify donors.

Verify recipients.

Suspend accounts.

Food Listings

Monitor platform activity.

Communities

Manage supported Chicago communities.

Future versions:

Manage additional cities.

Reports

Generate reports.

Export statistics.

Analytics

KPIs.

Donation trends.

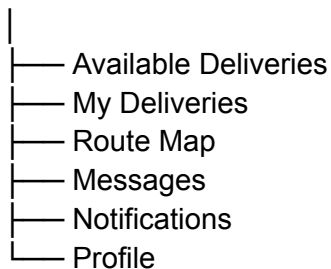
Community impact.

Growth metrics.

6.8 Future Volunteer Architecture

Although not included in the MVP, the platform architecture supports volunteers.

Dashboard



6.9 Navigation Hierarchy

The platform uses two navigation systems.

Public Website

Top Navigation

Desktop

↓

Hamburger Menu

Mobile

Dashboard

Sidebar

Desktop

↓

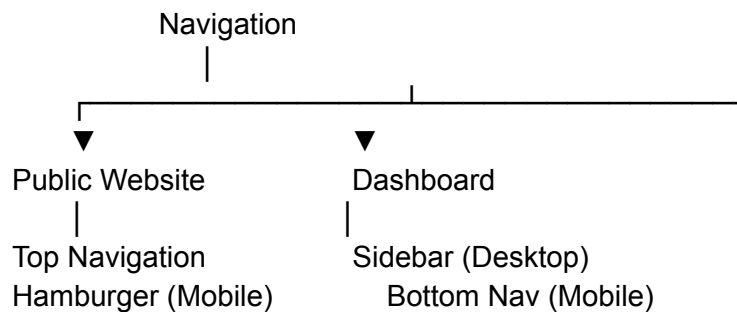
Bottom Navigation + Profile Menu

Mobile

This keeps the public experience simple while making operational tools easy to access for authenticated users.



Illustration 6.4 – Navigation Structure



6.10 Future Scalability

The Information Architecture has been intentionally designed to support future growth without requiring major redesign.

Future enhancements may include:

Additional Donors

- Restaurants
- Bakeries
- Farms
- Food Manufacturers

- Supermarkets
 - Community Organizations
-

Additional Recipients

- Soup Kitchens
 - School Meal Programs
 - Community Organizations
 - Regional Food Banks
-

Geographic Expansion

Current:

Chicago (6 Community Areas)

↓

Future:

All Chicago

↓

Illinois

↓

United States

↓

International Deployment

Additional Features

- AI-powered donation recommendations.
- Volunteer delivery coordination.
- Mobile applications.
- Real-time logistics tracking.
- Advanced reporting.

- Multi-language support.
-

6.11 Why This Architecture?

The chosen architecture directly reflects the research findings and project goals:

- **Clear separation of public and private areas** improves security and keeps operational data protected.
 - **Role-based dashboards** reduce clutter by showing users only the tools they need.
 - **Scalable structure** allows new organization types and locations to be added without redesigning the platform.
 - **Consistent navigation** makes the system easier to learn and use across different devices.
 - **Modular organization** enables future features—such as volunteer coordination or AI-assisted recommendations—to be integrated with minimal disruption.
-

Chapter Summary

This chapter defined the structural blueprint of the Surplus Food Redistribution Network. It organized the platform into public and private environments, described the page hierarchy for each user role, established the navigation structure, and demonstrated how the system can scale over time. This information architecture provides the foundation for the next chapter, where the navigation design and user interaction patterns will be refined into a consistent user experience.

CHAPTER SEVEN

Navigation Design

7.1 Introduction

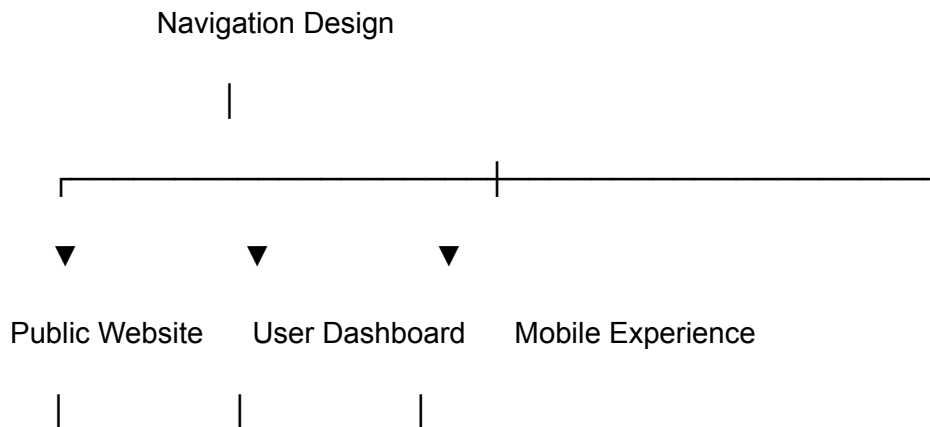
Navigation is one of the most critical aspects of user experience. Regardless of how powerful a platform is, users cannot benefit from its features if they cannot find them easily.

The navigation system for the **Surplus Food Redistribution Network** has been designed with the following objectives:

- Simplicity
- Consistency
- Accessibility
- Scalability
- Role-Based Navigation
- Mobile Responsiveness

Rather than using a single navigation style throughout the application, different navigation patterns are used depending on the user's context.

Illustration 7.1 – Navigation Philosophy



7.2 Navigation Goals

The navigation system was designed to achieve several objectives.

Goal 1 — Minimize User Effort

Users should reach their destination within **two or three clicks** whenever possible.

Goal 2 — Reduce Cognitive Load

Users should only see navigation items that are relevant to their role.

For example:

A grocery store manager does not need to see administrator tools.

Similarly,

A food pantry coordinator does not need access to donation creation pages.

Goal 3 — Maintain Consistency

Navigation components should remain consistent across every screen.

Examples include:

- Logo placement
 - Menu positioning
 - Button styling
 - Icons
 - Active page indicators
-

Goal 4 — Support Future Growth

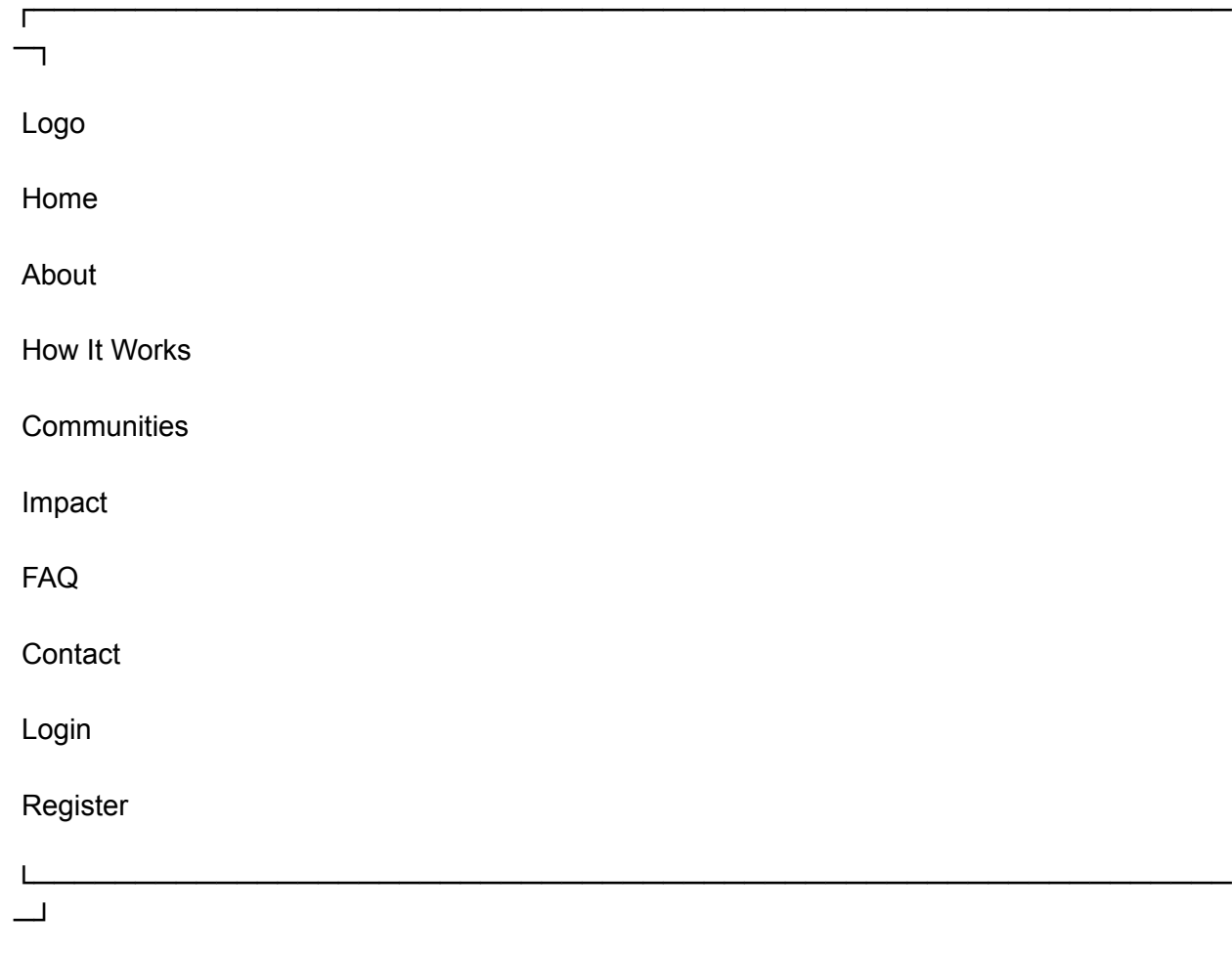
Additional pages should be added without redesigning the navigation system.

7.3 Public Website Navigation

The public website introduces the platform to visitors before they register.

Its navigation focuses on information rather than operational tasks.

Desktop Navigation



Why this layout?

The order follows a natural decision-making process:

Learn



Understand



Trust



Join

Visitors first learn about the platform before deciding to register.

Navigation Flow

Home



About



How It Works



Impact

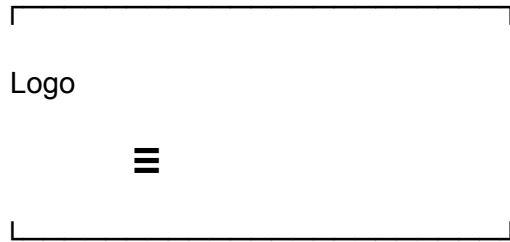


Register

7.4 Mobile Public Navigation

On mobile devices, screen space is limited.

Instead of displaying every navigation link horizontally, the platform uses a **hamburger menu**.



Selecting the menu icon opens:



Home

About

How It Works

Communities

Impact

FAQ

Contact

Login

Register

This approach keeps the interface uncluttered while ensuring all pages remain accessible.



Illustration 7.2 – Public Navigation

Desktop

Logo Home About Impact Login



Mobile

7.5 Dashboard Navigation Philosophy

Once authenticated, users transition from the **public website** to the **application dashboard**.

At this point, their priorities change.

Instead of learning about the platform, they now need quick access to operational tools.

For this reason, the dashboard uses a dedicated navigation system.

7.6 Desktop Dashboard Navigation

Desktop dashboards use a **left sidebar**.

This pattern is widely used in professional SaaS platforms because it provides:

- Clear organization
 - Easy expansion
 - Persistent navigation
 - Efficient access to tools
-

Example Sidebar

LOGO

Dashboard

Donate Food

My Donations

Pickup Schedule

Messages

Notifications

Profile

Settings

Logout

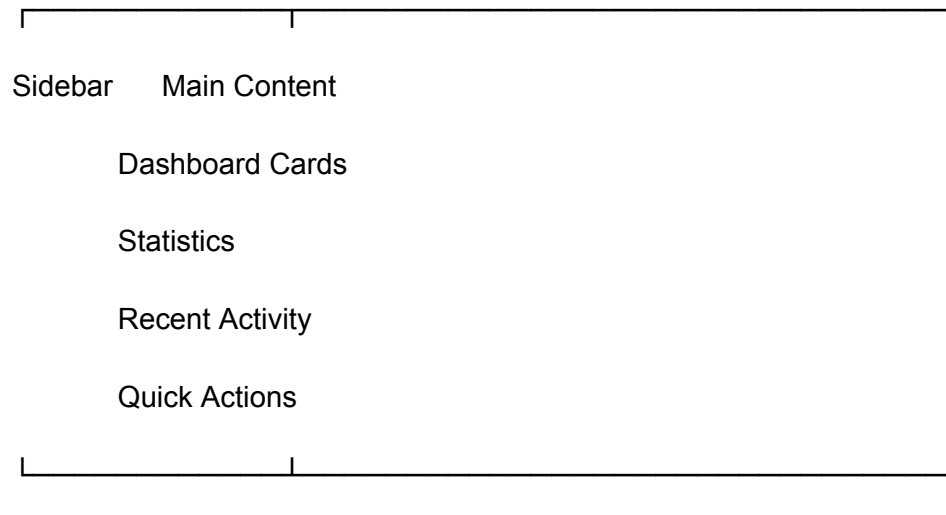
Why a Sidebar?

Compared to a top navigation bar, a sidebar:

- accommodates more menu items,
 - scales better as features grow,
 - reduces clutter,
 - allows users to navigate without leaving their current workflow.
-



Illustration 7.3 – Dashboard Layout



7.7 Mobile Dashboard Navigation

On mobile devices, a full sidebar would occupy too much space.

Instead, the platform uses a **bottom navigation bar** for primary actions and a **profile menu** for secondary actions.

Bottom Navigation



Home Food Map Chat Profile

Selecting **Profile** opens:

Profile

Settings

Notifications

Logout

This design allows one-handed operation while keeping frequently used features within easy reach.

Illustration 7.4 – Mobile Dashboard

Content



7.8 Role-Based Navigation

Each user role receives a navigation menu tailored to its responsibilities.

Donor Navigation

Dashboard

Donate Food

My Donations

Pickup Schedule

Messages

Notifications

Profile

Settings

Recipient Navigation

Dashboard

Available Food

My Reservations

Pickup Schedule

Messages

Notifications

Profile

Settings

Administrator Navigation

Dashboard

Users

Organizations

Food Listings

Communities

Analytics

Reports

Settings

Volunteer Navigation (Future)

Dashboard

Available Deliveries

Route Map

Messages

Profile



Illustration 7.5 – Role-Based Navigation

Login

↓

Authentication

↓

Donor

Recipient

Admin

Volunteer

↓

Each receives a unique dashboard menu

7.9 Navigation Behavior

To improve usability, the navigation follows several interaction rules:

- The current page is always highlighted.
- The platform logo returns users to the homepage or dashboard.
- Navigation remains visible while scrolling on desktop (sticky navigation).
- Sidebar items expand and collapse smoothly where applicable.
- On mobile, the bottom navigation remains fixed for quick access.

These behaviors create a predictable and intuitive experience.

7.10 Navigation and Scalability

The navigation system has been designed to accommodate future growth.

As the platform expands, additional items such as:

- Volunteer Management
- AI Recommendations
- Reports
- Multi-City Management
- Delivery Tracking

can be integrated into the sidebar without changing the overall navigation structure.

This flexibility ensures that the platform remains maintainable as new features are introduced.

7.11 Why This Navigation Design?

The chosen navigation strategy reflects the needs identified during the research and aligns with modern web application design practices.

Key reasons include:

- **Separation of public and private experiences** keeps marketing content distinct from operational workflows.
- **Role-based menus** reduce cognitive load by presenting only relevant options.
- **Responsive navigation** ensures usability across desktop, tablet, and mobile devices.
- **Sidebar navigation** supports feature-rich dashboards without overwhelming users.
- **Bottom navigation on mobile** enables efficient one-handed interaction for frequently used actions.

These decisions contribute to a navigation system that is simple, scalable, and user-centered.

Chapter Summary

This chapter defined the navigation strategy for both the public website and the authenticated platform. It introduced responsive navigation patterns, role-specific menus, desktop sidebars, mobile bottom navigation, and the guiding principles behind these design decisions. Together, these navigation patterns provide users with a consistent and efficient way to interact with the Surplus Food Redistribution Network.

CHAPTER EIGHT

Responsive Design

8.1 Introduction

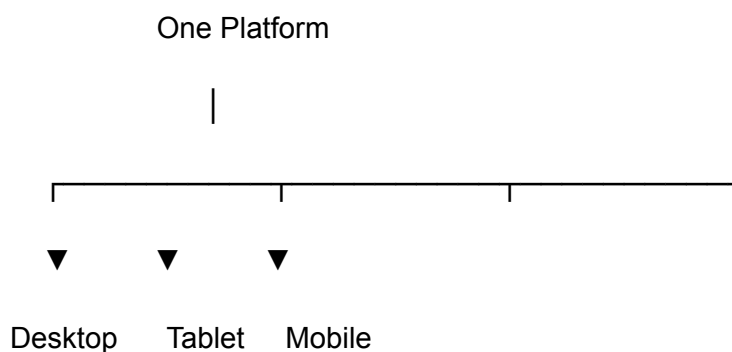
The **Surplus Food Redistribution Network** is designed to be accessible across a wide range of devices, including desktop computers, laptops, tablets, and smartphones. Because different users may access the platform in different environments, the interface must adapt without compromising usability, readability, or functionality.

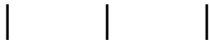
For example:

- A grocery store manager may create food donations from an office desktop.
- A food pantry coordinator may reserve donations from a tablet.
- An administrator may monitor platform activity from a laptop.
- A volunteer (future feature) may update delivery status using a smartphone.

Responsive design ensures that each of these users receives an experience optimized for their device.

Illustration 8.1 – Multi-Device Experience





Same Features, Optimized Layout

8.2 Responsive Design Goals

The responsive design strategy is guided by the following goals.

Goal 1 — Consistency

Users should immediately recognize the platform regardless of the device they use.

The layout may change, but the branding, colors, typography, icons, and interactions remain consistent.

Goal 2 — Usability

Interfaces should be optimized for the available screen size.

Buttons remain easy to tap.

Text remains readable.

Forms remain simple.

Navigation remains accessible.

Goal 3 — Performance

The platform should minimize unnecessary downloads and optimize images so that pages load efficiently, especially on slower mobile networks.

Goal 4 — Accessibility

Responsive layouts should support users with varying physical abilities by maintaining readable text sizes, sufficient spacing, and touch-friendly controls.

8.3 Device Categories

The platform is designed for four primary viewport categories.

Device	Width	Primary Users
Desktop	1200px and above	Administrators, Grocery Stores
Laptop	992px–1199px	Grocery Stores, Food Pantries
Tablet	768px–991px	Food Pantry Coordinators
Mobile	Below 768px	Coordinators, Future Volunteers

These breakpoints provide flexibility while remaining easy to maintain.

Illustration 8.2 – Responsive Breakpoints

Desktop

Laptop

Tablet

Mobile

8.4 Desktop Experience

The desktop interface provides the most complete experience and is intended for users performing complex tasks.

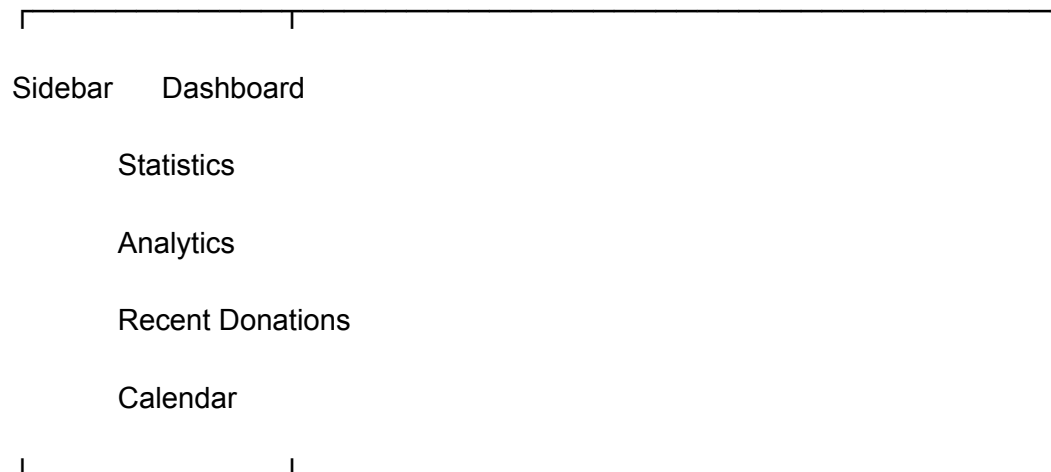
Typical users include:

- Grocery Store Managers
- Platform Administrators

Characteristics

- Full-width layout
 - Sidebar navigation
 - Multi-column dashboard
 - Data tables
 - Charts and analytics
 - Expanded forms
 - Persistent navigation
-

Desktop Dashboard Example



Desktop layouts prioritize visibility and productivity.

8.5 Tablet Experience

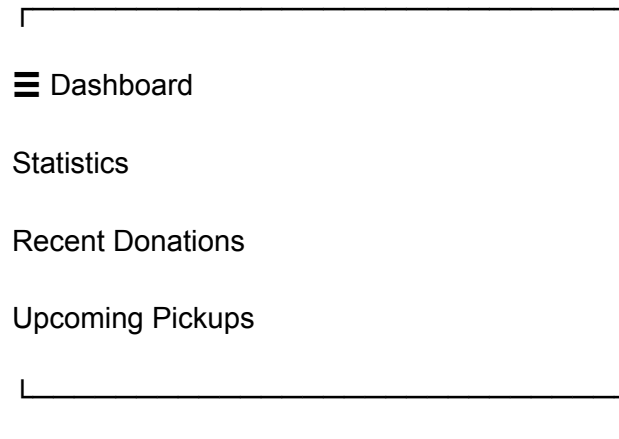
Tablet users often work in the field or within organization facilities.

The interface is adapted to medium-sized screens while retaining nearly all functionality.

Adjustments

- Smaller sidebar or collapsible navigation.
 - Two-column layouts where appropriate.
 - Larger touch targets.
 - Simplified dashboard cards.
-

Tablet Layout



8.6 Mobile Experience

Mobile users require a significantly different layout due to limited screen space.

Rather than shrinking the desktop interface, the platform reorganizes information to prioritize the most important tasks.

Typical mobile users include:

- Food Pantry Coordinators
 - Future Volunteers
 - Organization representatives on the move
-

Mobile Design Principles

- Single-column layouts.
- Large touch targets.

- Bottom navigation.
- Simplified forms.
- Collapsible sections.
- Reduced visual clutter.

Illustration 8.3 – Mobile Dashboard

Today's Donations

Pickup Today

Notifications



8.7 Responsive Navigation

Navigation changes depending on device size.

Public Website

Desktop

Top navigation with all menu items visible.

Tablet

Compact top navigation.

Mobile

Logo and hamburger menu.

Dashboard

Desktop

Permanent left sidebar.

Tablet

Collapsible sidebar.

Mobile

Bottom navigation with profile menu.



Illustration 8.4 – Navigation Evolution

Desktop

Logo Home About Impact Login

↓

Tablet

Logo Home About ☰

↓

Mobile

Logo ☰

8.8 Responsive Content Layout

Content presentation also adapts to screen size.

Desktop

- Three-column dashboard cards.
- Side-by-side forms.
- Large data tables.
- Map alongside listings.

Tablet

- Two-column layout.
- Stacked charts.
- Reduced spacing.

Mobile

- Single-column layout.
- Cards replace tables.
- Map opens on its own screen.
- Forms displayed one section at a time.



Illustration 8.5 – Content Adaptation

Desktop

Card Card Card

↓

Tablet

Card Card

Card

↓

Mobile

Card

Card

Card

8.9 Forms and Input Design

Food donation forms contain multiple fields, including:

- Food name.
- Quantity.
- Expiration date.
- Pickup window.
- Images.

Responsive adjustments include:

Desktop

Fields displayed side by side.

Tablet

Two-column grouping.

Mobile

Single-column layout with larger inputs and full-width buttons.

This improves readability and minimizes input errors.

8.10 Maps on Different Devices

The interactive map is a key feature of the platform.

Desktop

Displayed alongside food listings for simultaneous viewing.

Tablet

Occupies a larger portion of the screen while allowing quick switching to listings.

Mobile

Displayed as a dedicated screen with expandable information panels.

This approach prevents the map from becoming too small to use effectively.

Illustration 8.6 – Responsive Map

Desktop

Map | Listings

↓

Tablet

Large Map

↓

Mobile

Full-Screen Map

8.11 Dashboard Cards

Dashboard cards summarize key information.

Examples include:

- Active Donations
- Upcoming Pickups
- Reservations
- Notifications
- Community Impact

Desktop

Three to four cards per row.

Tablet

Two cards per row.

Mobile

One card per row.

This maintains readability while adapting to available screen space.

8.12 Why This Responsive Strategy?

The responsive design supports the project's objectives by ensuring that all users can interact with the platform regardless of the device they use.

Key benefits include:

- Improved usability across different environments.
- Reduced learning curve through consistent design.
- Better accessibility for users with varying technical abilities.
- Efficient use of screen space.
- Scalable layouts that accommodate future features.

Rather than simply resizing the interface, the design reorganizes content and navigation to provide the best possible experience on each device.

Chapter Summary

This chapter described how the Surplus Food Redistribution Network adapts to different screen sizes through responsive design. It outlined device categories, responsive navigation, content adaptation, form optimization, map behavior, and dashboard layouts. These decisions ensure that the platform remains functional, accessible, and user-friendly across desktops, tablets, and mobile devices.

Excellent. This chapter is where your frontend work starts to look like documentation from a professional product team. A **Design System** is much more than choosing colors—it defines the reusable visual rules that keep the entire platform consistent.

CHAPTER NINE

Design System

9.1 Introduction

A design system is a collection of reusable design standards, components, and guidelines that ensure consistency across the Surplus Food Redistribution Network.

Rather than designing each page independently, the platform follows a unified system of colors, typography, spacing, icons, buttons, forms, and layouts. This approach improves usability, simplifies development, and makes the platform easier to maintain as it grows.

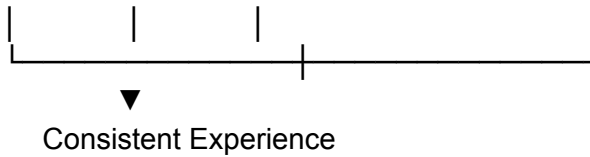
For the Surplus Food Redistribution Network, the design system was created with four priorities:

- Simplicity
- Accessibility
- Consistency
- Scalability

These principles help ensure that every page feels familiar to users while remaining flexible enough to support future features.

Illustration 9.1 – Design System Foundation





9.2 Design Principles

The visual identity of the platform is based on the following principles.

Simplicity

The interface avoids unnecessary visual complexity. Every element should have a clear purpose.

Clarity

Users should immediately understand what actions they can take and where to find important information.

Accessibility

Color contrast, readable typography, and keyboard navigation support users with different abilities.

Trust

The visual style should communicate reliability and professionalism, encouraging organizations to participate.

Scalability

The design should accommodate new features and user roles without requiring a complete redesign.

9.3 Color System

The platform uses a color palette inspired by sustainability, community, and trust.

Color	Purpose	Suggested Hex
Green	Primary actions, success, sustainability	#2E7D32
Blue	Navigation, links, trusted information	#1565C0
Orange	Warnings and time-sensitive actions	#F57C00
Red	Errors and destructive actions	#D32F2F
Gray	Borders, backgrounds, secondary text	#E0E0E0
White	Main background	#FFFFFF
Dark Gray	Headings and primary text	#263238

Why These Colors?

- **Green** reflects food sustainability and positive actions.
 - **Blue** conveys trust and reliability.
 - **Orange** draws attention to expiring food or pending actions.
 - **Red** highlights errors or critical alerts.
 - **Neutral grays** keep the interface clean and readable.
-

Illustration 9.2 – Color Roles

Green → Success / Donate
 Blue → Navigation / Links
 Orange → Pending / Warning
 Red → Error / Delete
 Gray → Neutral
 White → Background

9.4 Typography

Typography establishes a clear hierarchy and improves readability.

Recommended Font

Inter

Reasons:

- Excellent readability on screens.
 - Modern appearance.
 - Wide language support.
 - Popular in dashboards and SaaS products.
-

Type Scale

Element	Size
Page Title	36px
Section Heading	28px
Card Title	22px
Body Text	16px
Caption	14px
Small Labels	12px

Font Weights

Weight	Usage
Bold	Page titles
Semi-bold	Section headings
Medium	Buttons
Regular	Body content

9.5 Spacing System

Consistent spacing improves readability and visual organization.

An **8-point spacing system** is adopted throughout the platform.

Spacing	Value
XS	4px
Small	8px
Medium	16px
Large	24px
XL	32px
XXL	48px

Using a consistent spacing scale creates a balanced layout and simplifies frontend development.

9.6 Grid System

The platform uses a flexible grid layout.

Desktop

12-column grid.

Tablet

8-column grid.

Mobile

4-column grid.

This allows content to adapt smoothly to different screen sizes while maintaining alignment.

Illustration 9.3 – Responsive Grid

Desktop

□□□□□□□□□□

Tablet

□□□□□□□

Mobile

□□□□

9.7 Buttons

Buttons represent the primary way users interact with the platform.

Primary Button

Purpose:

- Donate Food
- Reserve Donation
- Save Changes
- Register

Appearance:

- Green background
 - White text
 - Rounded corners
-

Secondary Button

Purpose:

- Cancel
- Back
- Learn More

Appearance:

- White background

- Blue border
 - Blue text
-

Danger Button

Purpose:

- Delete
- Remove
- Cancel Reservation

Appearance:

- Red background
 - White text
-

Disabled Button

Purpose:

Prevent interaction when required information is missing.

Illustration 9.4 – Button Types

[Donate Food]

[Learn More]

[Delete]

[Disabled]

9.8 Form Design

Forms are central to the platform, especially for creating donations and managing organization profiles.

Form Guidelines

- Labels positioned above inputs.
- Clear placeholder text.
- Required fields clearly indicated.
- Validation messages displayed near the relevant field.
- Logical grouping of related information.

Example fields include:

- Food Name
 - Quantity
 - Expiration Date
 - Pickup Window
 - Description
 - Upload Images
-

9.9 Cards

Cards organize information into manageable sections.

Examples include:

- Donation summaries.
- Dashboard statistics.
- Organization profiles.
- Notifications.
- Community impact.

Each card includes:

- Title
- Key information
- Action button (if applicable)

Cards improve readability and responsiveness, especially on mobile devices.

Illustration 9.5 – Example Card

Fresh Vegetables

Available:
25 Boxes

Expires:
Tomorrow

[Reserve]

9.10 Icons

Icons provide quick visual recognition and reduce the need for excessive text.

Examples:

Icon	Meaning
	Dashboard
	Donations
	Map
	Notifications
	Messages
	Profile
	Settings

Icons are always accompanied by text labels to improve accessibility.

9.11 Status Indicators

The platform uses consistent status badges to communicate the state of donations and reservations.

Status	Color
Available	Green
Reserved	Blue
Pending Pickup	Orange
Completed	Gray
Expired	Red

These visual indicators help users quickly understand the current status of an item.

9.12 Data Tables

Administrators and donors often need to view large amounts of information.

Tables should include:

- Search
 - Sorting
 - Pagination
 - Filters
 - Responsive adaptation (cards on mobile)
-

9.13 Empty States

When no data is available, the platform provides clear guidance instead of blank pages.

Example:

No donations available yet.

Check back later or adjust your filters.

Empty states reduce confusion and encourage users to take appropriate actions.

9.14 Feedback and Notifications

The platform provides immediate feedback after user actions.

Examples:

- Donation successfully created.
- Reservation confirmed.
- Profile updated.
- Error while uploading image.

Feedback appears as temporary notification banners or toast messages.

9.15 Design Consistency

To maintain a cohesive experience:

- Colors remain consistent across all screens.
- Buttons use the same styles.
- Cards share identical spacing and typography.
- Icons follow a single visual style.
- Navigation behaves consistently.
- Forms use the same validation patterns.

A consistent design system reduces learning time for users and speeds up development for the team.

9.16 Why This Design System?

The chosen design system directly supports the project's goals by creating an interface that is:

- Easy to learn.
- Visually consistent.
- Accessible.
- Scalable.

- Professional.

It also ensures that future contributors can build new pages without creating inconsistencies, making the platform easier to maintain over time.

Chapter Summary

This chapter defined the visual language of the Surplus Food Redistribution Network. It established standards for colors, typography, spacing, grids, buttons, forms, cards, icons, and status indicators. Together, these elements form a reusable design system that promotes consistency, usability, and scalability across the platform.

Perfect. This is actually one of the most valuable chapters for a frontend developer because it bridges the gap between **design** and **development**. Instead of talking about pages, we define the reusable building blocks that make up every page.

Professional teams using React, Vue, Angular, or even plain HTML/CSS often start by building these reusable components.

CHAPTER TEN

Component Design

10.1 Introduction

The Surplus Food Redistribution Network is designed using a **component-based architecture**, where the user interface is constructed from reusable building blocks rather than designing each page independently.

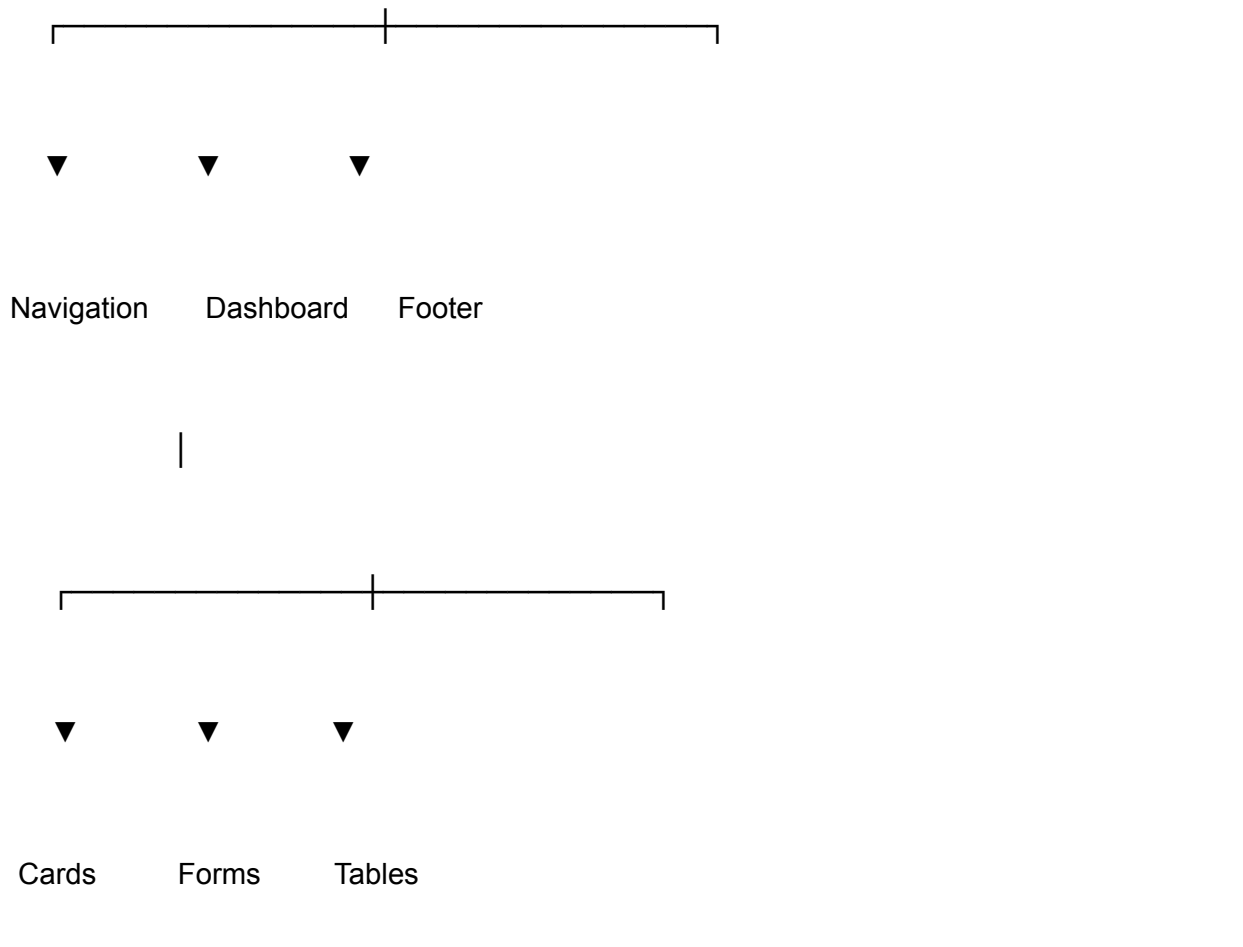
Each component has a specific responsibility and can be reused across multiple pages. This approach improves consistency, simplifies maintenance, reduces duplicated code, and accelerates future development.

The component library also supports scalability, allowing new features to be introduced without redesigning existing interfaces.

Illustration 10.1 – Component-Based Design

Web Page

|



10.2 Component Design Principles

Every component follows five principles.

Reusable

A component should be usable on multiple pages without modification.

Example:

A **Food Card** appears on:

- Dashboard
- Search Results
- Community Listings
- Favorites (Future)

Consistent

Buttons, typography, spacing, colors, and icons remain identical throughout the platform.

Responsive

Each component adapts automatically to desktop, tablet, and mobile layouts.

Accessible

Every interactive element supports:

- Keyboard navigation
 - Screen readers
 - High color contrast
 - Proper focus indicators
-

Modular

Each component performs one specific task.

10.3 Navigation Component

The Navigation Component is the user's primary method of moving throughout the platform.

Two navigation styles are used.

Public Website

Top Navigation

Includes:

- Home
 - About
 - How It Works
 - Communities
 - Impact
 - FAQ
 - Contact
 - Login
 - Register
-

Dashboard

Sidebar Navigation

Includes role-specific menu items.

Mobile

Hamburger Menu

↓

Bottom Navigation

Component Structure

Logo

Navigation Links

Search (Future)

User Profile

Notifications

Illustration 10.2 – Navigation Component

Logo

Home

About

Impact

Login

Register

10.4 Dashboard Card Component

Dashboard cards provide users with quick summaries of important information.

Examples include:

- Active Donations
- Reservations
- Pickups
- Notifications
- Community Impact

Card Structure

Icon

Title

Primary Number

Description

Action Button

Example

Active Donations

28

Food listings available today.

[View]

Why This Component?

Instead of reading long tables, users can quickly understand platform activity through visual summaries.

10.5 Food Listing Card

One of the most frequently used components.

Appears in:

- Search Results
 - Available Food
 - Dashboard
 - Community Listings
-

Card Information

Each food card displays:

- Food image
 - Food name
 - Donor organization
 - Quantity
 - Expiration date
 - Pickup location
 - Availability status
 - Reserve button
-

Illustration 10.3 – Food Card

 Fresh Vegetables

Donor:

Fresh Market

Quantity:

25 Boxes

Expires:

Tomorrow

Available

[Reserve]

10.6 Search Component

Users must quickly locate available food.

The search component allows filtering by:

- Food type
- Organization
- Community
- Date
- Availability

Search Layout

Search...

[Filter]

Results

Future enhancements may include:

- AI search suggestions.
 - Voice search.
 - Smart recommendations.
-

10.7 Filter Panel

The filter component refines search results.

Possible filters include:

Food Type

Organization

Distance

Expiration Date

Community Area

Availability

Illustration 10.4 – Filter Panel

Filters

☐ Vegetables

☐ Dairy

☐ Fruit

☐ Bakery

☐ Meat

Community

Availability

Apply

10.8 Interactive Map Component

The map is a central feature of the platform.

Its purpose is to display participating organizations geographically.

During the MVP, the map focuses on the six selected community areas in Chicago.

Future versions may expand to additional cities and regions.

Features

Display:

- Donor organizations
 - Recipient organizations
 - Community boundaries
 - Pickup locations (authenticated users)
-

Public visitors will only see participating organizations and service areas, while authenticated users can access operational details appropriate to their role.



Illustration 10.5 – Map Component

Map

 Grocery Store

 Food Pantry

10.9 Donation Form Component

This component allows donor organizations to publish surplus food.

Fields

Food Name

Category

Quantity

Expiration Date

Pickup Window

Description

Image Upload

Submit Button

Validation includes:

- Required fields.
 - Date validation.
 - Quantity validation.
 - Image format validation.
-

10.10 Reservation Component

Recipients use this component to reserve food donations.

Displays:

Donation Details

Availability

Pickup Time

Reserve Button

Confirmation Message

This component simplifies the reservation process while reducing communication delays.

10.11 Notification Component

Notifications keep users informed of important platform events.

Examples:

- Donation reserved.
 - Pickup reminder.
 - New donation available.
 - Account verification.
 - System announcements.
-

Notifications are grouped into:

Unread

Read

Archived

10.12 Messaging Component

The messaging component enables communication between participating organizations.

Supported interactions include:

- Donation clarification.
- Pickup coordination.

- Schedule updates.

For privacy and safety, communication is limited to verified organizations within the platform.

10.13 Data Table Component

Administrators require structured views of platform data.

Tables support:

- Search
 - Sorting
 - Pagination
 - Filtering
 - Export (future)
-

Typical tables include:

Users

Organizations

Food Listings

Reports

Community Statistics

10.14 Profile Component

Each organization maintains a profile containing:

Organization Name

Organization Type

Address

Contact Person

Phone Number

Email

Operating Hours

Verification Status

Updating profile information ensures accurate coordination between donors and recipients.

10.15 Modal Component

Modal dialogs are used for actions that require user confirmation or additional information.

Examples include:

- Confirm reservation
- Delete donation
- View donation details
- Edit profile
- Confirm logout

Using modals avoids unnecessary page navigation and keeps users focused on their current task.

10.16 Loading States

To improve the user experience during data retrieval, components display loading indicators.

Examples include:

- Skeleton cards while food listings load.
- Loading spinner during login.
- Progress indicator during image uploads.

These indicators reassure users that the system is processing their request.

10.17 Empty States

When no data is available, components provide helpful guidance rather than blank screens.

Examples:

No donations available in this community.

No upcoming pickups scheduled.

You have no unread notifications.

Empty states may also include actions such as:

- Browse all donations
- Refresh
- Create donation

10.18 Error States

When an operation fails, the interface communicates the problem clearly.

Examples:

- Failed to upload image.
- Network connection lost.
- Donation no longer available.
- Reservation could not be completed.

Each error message should explain the issue and, where possible, suggest the next step.

10.19 Component Relationships

The components work together to create a cohesive user experience.

Illustration 10.6 – Component Relationships

Navigation



Dashboard



Cards



Food Listings



Reservation



Notifications



Profile

This modular structure allows each part of the interface to evolve independently while maintaining consistency across the application.

10.20 Why This Component Design?

The component-based approach supports the project's objectives by:

- Promoting consistency across all screens.
- Reducing duplicated design and development effort.
- Simplifying maintenance and future enhancements.
- Supporting responsive layouts across devices.
- Allowing the frontend to scale as new features are introduced.

This approach aligns with modern frontend development practices and prepares the platform for future growth.

Chapter Summary

This chapter defined the reusable interface components that form the foundation of the Surplus Food Redistribution Network. By adopting a component-based architecture, the platform achieves consistency, maintainability, responsiveness, and scalability. These components provide the building blocks for all current and future screens within the application.

CHAPTER ELEVEN

Accessibility

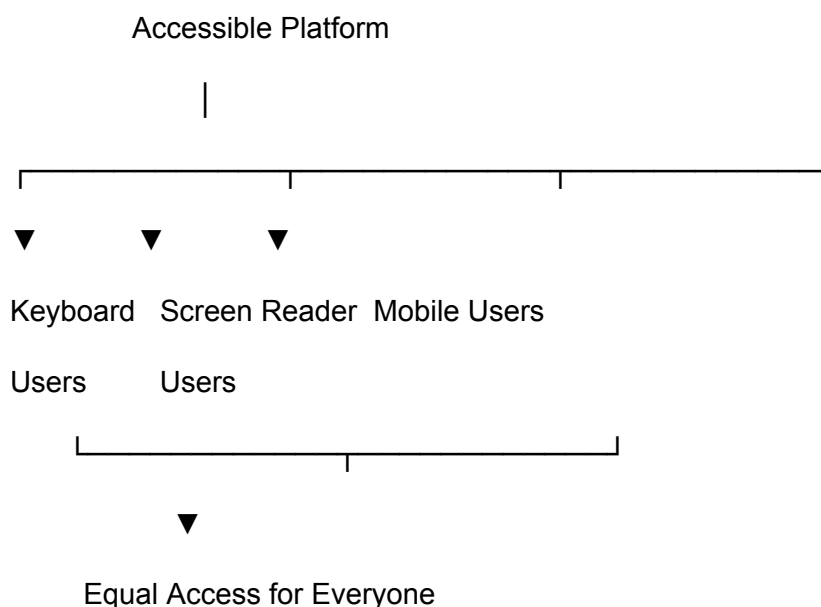
11.1 Introduction

Accessibility ensures that digital platforms can be used by the widest possible range of people, regardless of age, ability, or technical experience. The Surplus Food Redistribution Network is intended to serve grocery stores, food pantries, community organizations, and administrators with varying levels of digital literacy and different accessibility needs.

The platform therefore adopts accessibility principles that improve usability for everyone, not only users with disabilities.

Designing with accessibility in mind also supports the project's broader mission of creating an inclusive and equitable solution for food redistribution.

Illustration 11.1 – Inclusive Design



11.2 Accessibility Goals

The accessibility strategy is guided by four key goals.

Goal 1 — Perceivable

Information should be presented in ways that all users can perceive.

Examples:

- Sufficient color contrast.
- Readable typography.
- Descriptive labels.
- Alternative text for images.

Goal 2 — Operable

Users should be able to navigate and interact with the platform using different input methods.

Examples:

- Keyboard navigation.
- Large touch targets.
- Logical tab order.
- Visible focus indicators.

Goal 3 — Understandable

The interface should use clear language and predictable interactions.

Examples:

- Consistent navigation.
 - Simple labels.
 - Helpful error messages.
 - Familiar icons.
-

Goal 4 — Robust

The platform should work reliably across modern browsers, devices, and assistive technologies.

11.3 Readable Typography

Readable text is essential for all users.

The platform follows these typography guidelines:

- Minimum body text size of **16px**.
- Clear heading hierarchy.
- Adequate line spacing.
- Left-aligned paragraphs for easier reading.
- Consistent font usage across all pages.

These choices reduce eye strain and improve comprehension.

11.4 Color Contrast

Color should never be the only way information is communicated.

The platform uses high-contrast combinations to ensure readability.

Examples:

Element	Example
White background	Dark gray text
Green buttons	White text

Error messages Red icon + descriptive text

Success
messages Green icon + descriptive text

Status indicators include both **color** and **text labels**.

Example:

 Available

 Pending Pickup

 Expired

This helps users with color vision deficiencies distinguish different states.

Illustration 11.2 – Status Indicators

 Available

 Pending

 Expired

Each includes both color and text.

11.5 Keyboard Navigation

Some users rely entirely on a keyboard to navigate web applications.

The platform supports:

- Tab navigation.
- Shift + Tab to move backwards.
- Enter to activate buttons.
- Escape to close dialogs.
- Arrow keys where appropriate.

Interactive elements receive a visible focus outline when selected.

11.6 Accessible Forms

Forms are one of the most important interactions on the platform.

To improve accessibility:

- Every input has a visible label.
- Required fields are clearly identified.
- Instructions are provided before submission.
- Validation messages appear next to the relevant field.
- Error messages explain how to fix the problem.

Example:

 *Expiration date cannot be earlier than today.*

This is more helpful than simply displaying "Invalid input."

11.7 Alternative Text for Images

Images used throughout the platform include descriptive alternative (alt) text.

Examples:

- "Photo of fresh vegetables available for donation."
- "Map showing participating grocery stores."
- "Organization profile image."

Alternative text enables screen readers to describe images for visually impaired users.

11.8 Responsive Accessibility

Accessibility is maintained across all supported devices.

Desktop

- Full keyboard support.
- Large clickable buttons.
- Clear navigation.

Tablet

- Touch-friendly spacing.
- Simplified layouts.

Mobile

- Large tap targets.
- Bottom navigation.
- Single-column forms.
- Readable text without zooming.

Users should not need to pinch-to-zoom to interact with the interface.

11.9 Error Prevention

The platform helps users avoid mistakes before they occur.

Examples include:

- Preventing submission of incomplete forms.
- Validating expiration dates.
- Confirming destructive actions.
- Highlighting missing required fields.

Where possible, the interface guides users toward successful completion rather than simply reporting errors afterward.

11.10 Notifications and Feedback

System feedback should be immediate and understandable.

Examples:

✓ Donation successfully published.

✓ Reservation confirmed.

⚠ Pickup time updated.

✗ Image upload failed. Please try again.

Notifications are concise, easy to dismiss, and do not interrupt the user's workflow unnecessarily.

11.11 Language and Content

The platform uses simple, professional language that can be understood by users with different technical backgrounds.

Guidelines include:

- Avoid technical jargon where possible.
 - Use clear action labels such as "Reserve Donation" instead of vague terms like "Submit."
 - Write concise instructions.
 - Maintain consistent terminology throughout the platform.
-

11.12 Accessibility in Maps

The interactive map includes accessibility considerations such as:

- Zoom controls.

- Keyboard-accessible controls (where supported by the chosen map library).
- Search functionality for users who prefer not to navigate visually.
- Text-based organization listings as an alternative to relying solely on the map.

This ensures users can access location information even if interacting with the map itself is difficult.

11.13 Accessibility Testing

Before deployment, the platform should be evaluated using accessibility testing methods.

Suggested checks include:

- Keyboard-only navigation.
- Screen reader testing.
- Color contrast evaluation.
- Responsive testing on multiple devices.
- Form validation testing.
- Browser compatibility testing.

Testing helps identify issues that may not be apparent during development.



Illustration 11.3 – Accessibility Checklist

☒ Readable text

☒ Keyboard navigation

☒ High contrast

☒ Alternative text

☑ Accessible forms

☑ Responsive layouts

☑ Helpful error messages

11.14 Why Accessibility Matters

Accessibility directly supports the objectives of the Surplus Food Redistribution Network by ensuring that organizations of different sizes, technical abilities, and accessibility needs can participate effectively.

Benefits include:

- Broader user participation.
- Improved usability for all users.
- Greater inclusivity.
- Reduced user frustration.
- Compliance with widely accepted accessibility principles.

By considering accessibility from the beginning of the design process, the platform becomes more resilient, easier to use, and better aligned with its mission of serving communities.

Chapter Summary

This chapter outlined the accessibility strategy for the Surplus Food Redistribution Network. It described how readable typography, accessible forms, keyboard navigation, high-contrast colors, alternative text, responsive layouts, and inclusive language contribute to a platform that can be used effectively by a wide range of users. Accessibility was treated as a core design principle rather than an optional enhancement.

CHAPTER TWELVE

User Flows

12.1 Introduction

User flows describe the sequence of steps a user takes to accomplish a specific goal within the platform. They help ensure that every interaction is logical, efficient, and aligned with the needs identified during the research phase.

For the Surplus Food Redistribution Network, user flows were designed around the primary activities of each user role. These flows minimize unnecessary steps while ensuring that food donations can be created, discovered, reserved, and collected efficiently.

The main user flows include:

- Public Visitor Registration
 - Donor Donation Flow
 - Recipient Reservation Flow
 - Pickup Coordination Flow
 - Administrator Verification Flow
 - Future Volunteer Delivery Flow
-

Illustration 12.1 – User Flow Overview

User Flows

|



Visitor Donor Recipient Admin



Volunteer (Future)

12.2 Public Visitor Registration Flow

Objective

Allow eligible organizations to learn about the platform and register for an account.

Flow

Visit Website



Read About Platform



Learn How It Works



View Community Impact



Click Register



Choose Organization Type



Complete Registration Form



Submit Application



Await Verification



Receive Account Approval



Login

Why this flow?

Most organizations will not immediately create an account. They first need to understand:

- What the platform does.
- Whether they are eligible.
- How food redistribution works.

Providing information before registration builds trust and encourages informed participation.



Illustration 12.2 – Visitor Journey

Homepage



About

↓

How It Works

↓

Register

↓

Verification

↓

Dashboard

12.3 Donor Flow – Creating a Food Donation

Objective

Allow donor organizations to publish surplus food quickly and accurately.

Flow

Login



Donor Dashboard



Click "Donate Food"



Complete Donation Form



Upload Food Image



Enter Quantity



Select Pickup Window

↓

Submit Donation

↓

Donation Published

↓

Recipients Notified

Key Design Decisions

- A dedicated **"Donate Food"** button is prominently displayed on the dashboard.
- The donation form groups related fields together.
- Validation prevents incomplete or invalid submissions.
- Notifications are sent automatically after publication.

These decisions reduce the time required to create a donation and help prevent food from expiring before it is listed.

Illustration 12.3 – Donation Flow

Dashboard

↓

Donate Food

↓

Food Form

↓

Publish

↓

Available Donation

↓

Recipients Notified

12.4 Recipient Flow – Finding and Reserving Food

Objective

Enable recipient organizations to locate available food and reserve it before it becomes unavailable.

Flow

Login



Recipient Dashboard



Browse Available Food



Use Search / Filters



View Food Details



Click Reserve

↓

Confirm Reservation

↓

Pickup Scheduled

↓

Reservation Complete

Design Considerations

Recipients need to respond quickly because food availability is time-sensitive.

The interface therefore prioritizes:

- Search functionality.
- Filters.
- Status badges.
- Reserve button.
- Expiration date visibility.

These features help recipients make quick decisions.

Illustration 12.4 – Reservation Flow

Dashboard

↓

Available Food

↓

Food Details

↓

Reserve

↓

Confirmation

↓

Pickup

12.5 Pickup Coordination Flow

Objective

Coordinate food collection between donor and recipient organizations.

Flow

Donation Reserved



Pickup Time Confirmed



Reminder Notification



Recipient Arrives



Food Collected



Donation Completed

↓

Statistics Updated

Why this flow?

Without a structured pickup process:

- Food may remain uncollected.
- Organizations may experience scheduling conflicts.
- Food waste could still occur.

This workflow ensures accountability and improves coordination.

Illustration 12.5 – Pickup Flow

Reservation

↓

Pickup Schedule

↓

Reminder

↓

Collect Food

↓

Completed

12.6 Administrator Verification Flow

Objective

Ensure that only legitimate organizations gain access to the operational platform.

Flow

Organization Registers

↓

Application Submitted

↓

Administrator Reviews

↓

Verify Information



Approve

or

Reject



Email Notification



Account Activated

Why Verification?

Verification helps:

- Prevent fraudulent registrations.
 - Protect participating organizations.
 - Improve trust.
 - Maintain data quality.
-

Illustration 12.6 – Verification Flow

Register



Review



Approve



Platform Access

12.7 Future Volunteer Flow

Although volunteers are not included in the MVP, the platform is designed to support them in future versions.

Flow

Login



Available Deliveries



Accept Delivery



View Route



Collect Food



Deliver Food



Mark Complete

This workflow enables volunteers to assist when donor and recipient organizations require transportation support.

12.8 Error Handling Flows

User flows also account for situations where an action cannot be completed successfully.

Example 1 – Donation Form Validation

Submit Donation

↓

Validation Error

↓

Display Error Message

↓

Correct Information

↓

Submit Again

Example 2 – Food Already Reserved

Reserve Donation



Already Reserved



Notification Displayed



Return to Search Results

Example 3 – Network Failure

Submit Request



Connection Lost



Retry

↓

Success

or

Try Again Later

These recovery paths improve usability and reduce frustration.

12.9 User Flow Optimization

The platform minimizes unnecessary steps by:

- Providing clear calls to action.
- Displaying frequently used features prominently.
- Reducing the number of clicks required for common tasks.
- Using notifications to keep users informed.
- Keeping workflows consistent across user roles.

For example, a donor can create a new donation directly from the dashboard instead of navigating through multiple menus.

12.10 Relationship Between User Flows and Research

Each user flow directly addresses challenges identified in the research conducted by the data analyst.

Research Finding

User Flow Response

Food expires before collection

Fast donation creation and notifications

Communication delays

Built-in messaging and reminders

Limited visibility of available food

Search, filters, and interactive map

Coordination challenges

Structured reservation and pickup workflows

Need for trust

Organization verification before access

This demonstrates that the platform design is evidence-based rather than driven by assumptions.

12.11 Why These User Flows?

The user flows were developed to:

- Simplify common tasks.
- Reduce food redistribution delays.
- Improve coordination between organizations.
- Support users with varying technical experience.
- Align the interface with real operational workflows identified during the research.

By documenting these flows before implementation, the team can ensure that development remains focused on solving real user problems.

Chapter Summary

This chapter documented the primary user journeys within the Surplus Food Redistribution Network. It described how public visitors register, donors create food donations, recipients reserve available food, administrators verify organizations, and future volunteers support deliveries. The documented flows provide a clear blueprint for implementing the platform while ensuring that every interaction addresses a need identified in the research.

Perfect. This section is the **core of your actual application**. If someone gave this document to a frontend developer, they should be able to build almost the entire interface without asking many questions.

CHAPTER THIRTEEN (Part B)

Dashboard Screens

Unlike the Public Website, the Dashboard is only accessible to verified organizations and administrators. Every dashboard is role-based, meaning users only see features that are relevant to their responsibilities.

13.11 Donor Dashboard

Purpose

The Donor Dashboard is the primary workspace for organizations donating surplus food. It provides a quick overview of donation activity and allows users to create and manage food listings.

The dashboard is designed to minimize the time required to publish donations, helping reduce food waste.

Primary Users

- Grocery Stores
 - Bakeries
 - Restaurants
 - Supermarkets
 - Farms
 - Community Organizations (Donors)
-

Components

- Sidebar Navigation
 - Welcome Header
 - Dashboard Statistics
 - Quick Action Buttons
 - Recent Donations
 - Upcoming Pickups
 - Notifications
 - Community Activity
-

Suggested Layout

Sidebar

Dashboard

Donate Food

My Donations

Messages

Notifications

Profile

Welcome Mercy!

[Active Donations]

[Scheduled Pickups]

[Reservations]

[Community Impact]

Recent Donations

Upcoming Pickups

Dashboard Statistics

Display summary cards such as:

- Active Donations
- Donations Completed
- Food Saved
- Organizations Helped

These provide immediate insight into the donor's impact.

Quick Actions

The dashboard includes prominent buttons for frequently used tasks.

Examples:

- Donate Food
- View Donations
- View Pickup Schedule

This reduces the number of clicks needed for common actions.

Why This Screen Exists

Research showed that food often expires because donors lack a fast and organized method to publish available surplus.

The dashboard places donation creation at the center of the experience.

13.12 Donate Food Screen

Purpose

Enable donor organizations to create new food listings quickly and accurately.

Components

- Food Image Upload
 - Food Name
 - Category
 - Quantity
 - Unit
 - Expiration Date
 - Pickup Window
 - Pickup Instructions
 - Description
 - Submit Button
-

Layout

Food Image

Food Name

Category

Quantity

Expiration Date

Pickup Time

Description

[Publish Donation]

Validation

The form checks:

- Required fields
- Valid dates
- Positive quantities
- Image format

Errors are shown beside the relevant field.

Why This Screen Exists

The simpler the donation process, the more likely organizations are to publish food before it expires.

13.13 My Donations Screen

Purpose

Allow donors to manage previously published donations.

Components

Each donation card displays:

- Food Name
- Quantity
- Status
- Expiration Date
- Reservation Status

- Edit Button
 - Delete Button
-

Filters

Users can filter by:

- Available
 - Reserved
 - Completed
 - Expired
-

Actions

Users may:

- Edit donation
 - Delete donation
 - Mark as unavailable
 - View reservation details
-

Why This Screen Exists

Donors need complete visibility into their active and historical donations.

13.14 Recipient Dashboard

Purpose

Provide recipient organizations with a workspace focused on finding and reserving available food.

Primary Users

- Food Pantries
 - Soup Kitchens
 - School Meal Programs
 - Community Organizations (Recipients)
-

Components

- Dashboard Statistics
 - Search Bar
 - Community Filter
 - Recent Donations
 - Reservation History
 - Notifications
-

Suggested Layout

Welcome Organization

Search Donations

Filters

Available Donations

Recent Reservations

Upcoming Pickups

Statistics

Examples:

- Available Donations
- Active Reservations
- Food Collected
- Partner Organizations

Why This Screen Exists

Recipients need immediate access to available food because donations are time-sensitive.

13.15 Available Food Screen

Purpose

Display all food currently available for reservation.

This is one of the most frequently visited pages in the platform.

Components

- Search Bar
 - Filter Panel
 - Food Cards
 - Interactive Map Toggle
 - Sort Options
-

Food Card

Each listing contains:

- Image
 - Food Name
 - Donor Organization
 - Community Area
 - Quantity
 - Expiration Date
 - Reserve Button
-

Example

Fresh Bread

Bakery House

Near West Side

15 Boxes

Expires Tomorrow

[Reserve]

Filters

Food Type

Community Area

Organization

Expiration Date

Availability

Why This Screen Exists

Without centralized visibility, recipient organizations struggle to discover available food before it expires.

13.16 Food Details Screen

Purpose

Provide complete information about a selected donation.

Components

Large Food Image

Description

Donor Information

Pickup Window

Quantity

Location

Reserve Button

User Actions

Users can:

- Review donation details
 - View pickup information
 - Reserve food
 - Contact donor (if necessary)
-

Why This Screen Exists

Recipients need sufficient information before committing to a reservation.

13.17 Reservation Screen

Purpose

Confirm a recipient's reservation before final submission.

Components

Donation Summary

Pickup Date

Organization Information

Confirmation Checkbox

Reserve Button

Flow

Review

↓

Confirm

↓

Reservation Successful

Why This Screen Exists

Prevents accidental reservations and ensures users review important details.

13.18 Pickup Schedule Screen

Purpose

Display upcoming food collections.

Components

Upcoming Pickups

Calendar View

Status

Contact Information

Directions (Future)

Status

- Scheduled
 - Completed
 - Cancelled
-

Why This Screen Exists

Keeps organizations organized and reduces missed pickups.

13.19 Messages Screen

Purpose

Enable communication between verified organizations.

Components

Conversation List

Chat Window

Message Input

Attachment Support (Future)

Use Cases

Clarify pickup time.

Ask about food quantity.

Confirm availability.

Coordinate special requests.

Why This Screen Exists

Research identified communication delays as a major challenge in food redistribution.

13.20 Notifications Screen

Purpose

Keep users informed of important events.

Notification Types

New Donation

Reservation Confirmed

Pickup Reminder

Account Approved

System Announcement

Actions

Users can:

- Mark as read
 - Delete notification
 - Open related page
-

Why This Screen Exists

Timely notifications improve coordination and reduce missed opportunities.

13.21 Profile Screen

Purpose

Allow organizations to manage their account information.

Components

Organization Logo

Organization Name

Contact Person

Email

Phone

Address

Operating Hours

Save Changes Button

Why This Screen Exists

Accurate organization information improves trust and coordination.

13.22 Settings Screen

Purpose

Allow users to customize account preferences.

Settings Include

- Password
 - Notification Preferences
 - Language (Future)
 - Privacy Settings
 - Account Security
-

Why This Screen Exists

Gives users control over their account while supporting long-term platform maintenance.

13.23 Administrator Dashboard

Purpose

Provide administrators with a complete overview of platform activity and management tools.

Unlike donor and recipient dashboards, this dashboard focuses on monitoring, moderation, and reporting rather than daily food transactions.

Components

- Total Organizations
 - Pending Verifications
 - Active Donations
 - Completed Pickups
 - Community Statistics
 - Recent Platform Activity
-

Administrator Actions

Administrators can:

- Approve or reject organizations.
 - Review reported issues.
 - Monitor donation activity.
 - View platform-wide analytics.
 - Manage communities.
-

Why This Screen Exists

A central administrative interface ensures the platform remains secure, organized, and trustworthy.

13.24 Organization Management Screen

Purpose

Allow administrators to manage registered organizations.

Components

- Organization List
 - Search
 - Filters
 - Verification Status
 - View Details
 - Approve
 - Reject
 - Suspend (Future)
-

Why This Screen Exists

Organization verification is essential for maintaining trust and preventing misuse of the platform.

13.25 Community Management Screen

Purpose

Manage the six Chicago community areas included in the MVP.

Administrators can:

- View participating organizations.
 - Monitor community activity.
 - Track donation statistics.
 - Prepare the platform for expansion into additional cities in future versions.
-

Why This Screen Exists

The platform begins with six community areas, but the structure is intentionally scalable so new communities, cities, and regions can be added without redesigning the system.

13.26 Analytics & Reports Screen

Purpose

Provide administrators with insights into platform performance.

Metrics

- Total Donations
 - Food Saved
 - Meals Distributed (Estimated)
 - Active Organizations
 - Reservations Completed
 - Community Performance
-

Visualizations

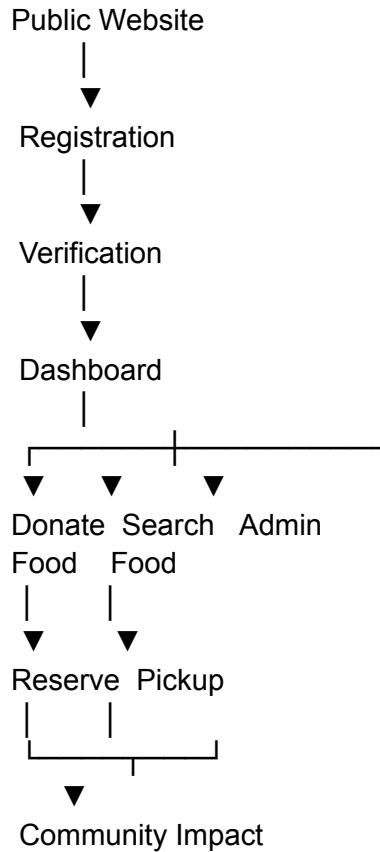
- Bar Charts
 - Line Charts
 - Pie Charts
 - Monthly Trends
 - Community Comparison
-

Why This Screen Exists

Analytics support evidence-based decision-making and demonstrate the platform's impact to stakeholders.

13.27 Screen Relationships

The screens are connected to support the complete lifecycle of food redistribution.



13.28 Why These Screens?

Every screen included in the platform exists because it addresses a challenge identified during your team's research:

Research Finding	Screen That Addresses It
Organizations are unaware of available food	Home Page, Communities Page, Available Food
Food expires before it can be redistributed	Donor Dashboard, Donate Food Screen
Recipients struggle to find donations	Search, Filters, Food Details

Communication delays affect coordination

Messages, Notifications

Lack of trust between organizations

Registration, Verification, Organization Management

Administrators need oversight

Admin Dashboard, Analytics, Community Management

This ensures the interface is driven by **user needs and research findings**, rather than by arbitrary feature selection.

Chapter Summary

This chapter described every major screen within the Surplus Food Redistribution Network, from the public-facing website to the authenticated dashboards used by donors, recipients, and administrators. Each screen was designed with a clear purpose and directly supports the workflows and research findings identified earlier in the project. Together, these screens provide a complete blueprint for implementing the frontend of the platform.

CHAPTER FOURTEEN

Frontend Technology Stack

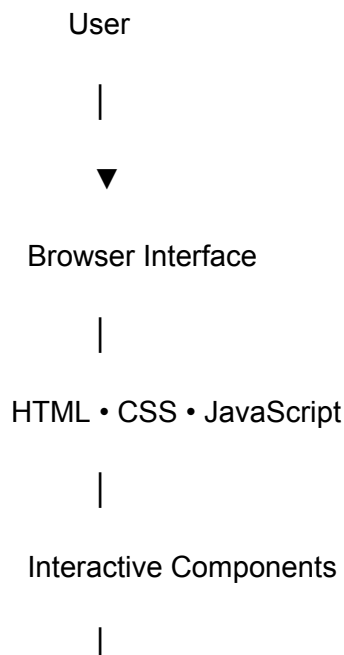
14.1 Introduction

The frontend of the **Surplus Food Redistribution Network** is responsible for presenting information, collecting user input, and enabling interaction between organizations participating in the food redistribution process.

The selected technology stack emphasizes accessibility, maintainability, responsiveness, and ease of future expansion. Since this project is developed as part of the BUILD program, preference is given to widely adopted, well-documented, and free technologies.

The technology choices also support future migration from a prototype into a production-ready platform.

Illustration 14.1 – Frontend Architecture



14.2 Frontend Development Goals

The frontend is designed to achieve several objectives:

- Provide an intuitive user experience.
 - Ensure compatibility across desktop and mobile devices.
 - Support accessibility best practices.
 - Minimize loading times.
 - Present complex information in a clear and organized manner.
 - Allow future expansion without redesigning the interface.
-

14.3 Core Technologies

HTML5

HTML provides the structural foundation of every page within the platform.

It is responsible for organizing content such as:

- Navigation menus
- Forms
- Tables
- Cards
- Dashboards
- Buttons
- Images

Using semantic HTML improves accessibility and helps search engines better understand page structure.

CSS3

CSS controls the visual presentation of the platform.

It is used for:

- Responsive layouts
- Color themes
- Typography
- Grid systems
- Animations
- Card layouts
- Navigation styling

A consistent CSS architecture ensures the interface remains visually cohesive across all pages.

JavaScript

JavaScript provides interactivity throughout the platform.

Examples include:

- Form validation
- Dynamic dashboards
- Interactive maps
- Notifications
- Search
- Filters
- Reservation workflows
- User authentication feedback

JavaScript enables the application to respond immediately to user actions without requiring full page reloads.

14.4 Responsive Layout Strategy

The interface follows a **mobile-first** responsive approach.

Layouts automatically adjust for:

- Desktop
- Laptop
- Tablet
- Mobile

Responsive behavior includes:

- Flexible grids.
 - Scalable images.
 - Collapsible navigation.
 - Adaptive dashboard layouts.
 - Touch-friendly controls.
-

14.5 Mapping Technology

An interactive map is one of the platform's core features.

For the MVP, the map focuses on the **six selected Chicago community areas** while keeping the design flexible enough to support future expansion into other cities.

The map should support:

- Displaying participating organizations.
- Viewing community coverage.
- Searching locations.
- Displaying organization markers.

A free mapping solution should be used to ensure there are no licensing costs during development.

14.6 UI Components

The frontend uses reusable components to improve consistency.

Examples include:

- Navigation Bar
- Sidebar
- Dashboard Cards
- Food Cards
- Search Bar
- Filter Panel
- Forms
- Tables

- Notifications
- Modals

Building reusable components reduces duplicated code and simplifies maintenance.

14.7 Frontend Performance

Performance is an important aspect of user experience.

Optimization strategies include:

- Compressing images before upload.
- Lazy loading images where appropriate.
- Minimizing CSS and JavaScript files.
- Avoiding unnecessary network requests.
- Reusing components instead of duplicating them.

These techniques help pages load quickly, especially on slower internet connections.

14.8 Browser Compatibility

The platform should function correctly on modern browsers including:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari

Testing across multiple browsers helps ensure a consistent experience for all users.

14.9 Security Considerations (Frontend)

Although most security measures belong to the backend, the frontend also contributes by:

- Validating form input.

- Preventing accidental duplicate submissions.
- Displaying clear authentication feedback.
- Managing user sessions securely.
- Avoiding exposure of sensitive information in the interface.

These measures improve both usability and overall system reliability.

14.10 Maintainability

To support future development, the frontend should be organized into reusable modules.

Examples include:

components/

- Navbar
- Sidebar
- FoodCard
- DashboardCard
- SearchBar
- Footer

pages/

- Home
- About
- Dashboard
- DonateFood
- Communities

This structure makes it easier for multiple developers to work on the project simultaneously.

14.11 Why This Technology Stack?

The selected frontend technologies support the project's objectives because they are:

- Free to use.
- Widely adopted.
- Easy to learn.
- Well documented.

- Compatible with modern browsers.
- Suitable for responsive design.
- Scalable for future enhancements.

These characteristics make them appropriate for an academic project while also providing a strong foundation for future development.

Chapter Summary

This chapter described the proposed frontend technology stack for the Surplus Food Redistribution Network. It outlined the role of HTML, CSS, and JavaScript, discussed responsive design strategies, reusable components, performance considerations, browser compatibility, and frontend maintainability. The selected technologies support the project's goals of usability, scalability, and accessibility while remaining practical for implementation within the BUILD project.

Perfect. I actually agree with the direction we're taking. Since this is for the **Google BUILD Capstone**, we should make it read like a professional **Software Design Document (SDD)** rather than just a UI description.

The next chapter should connect the frontend to the backend. This is something many student projects forget, but professional teams always define it.

CHAPTER FIFTEEN

Frontend Data Requirements

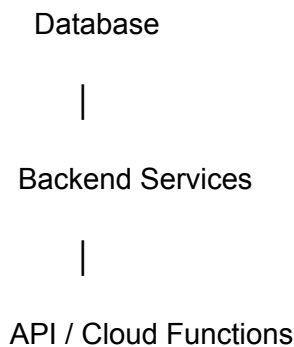
15.1 Introduction

Every screen within the Surplus Food Redistribution Network depends on accurate and up-to-date information. While the frontend is responsible for displaying this information, the backend is responsible for storing, processing, and returning it.

This chapter defines the data required by each screen, the user roles permitted to access it, and the actions that can be performed. Clearly documenting these requirements helps ensure that frontend and backend development remain aligned throughout the project.



Illustration 15.1 – Frontend Data Flow



|
Frontend Screens

|
Users

15.2 Home Page Data

Purpose

The Home Page displays general information that is publicly accessible.

Data Displayed

- Platform name
- Mission statement
- Hero banner
- Number of participating organizations
- Communities served
- Estimated meals redistributed
- Success stories
- Footer information

Access

Public

Editable By

Administrator

15.3 Communities Page

This page introduces visitors to the supported service areas.

Data Displayed

Community Name

Community Description

Organization Count

Map Boundary

Community Statistics

Community Image

Access

Public

Editable By

Administrator

15.4 Organization Directory

Rather than displaying every registered user, the public platform presents verified participating organizations.

Data Displayed

Organization Name

Organization Type

Community

Logo

Verification Status

Description

Visible To

Public Visitors

Editable By

Administrator

Verified Organization

15.5 Donor Dashboard

The donor dashboard displays information specific to the authenticated donor.

Dashboard Statistics

Active Donations

Completed Donations

Scheduled Pickups

Food Saved

Unread Notifications

Recent Donations

Food Name

Quantity

Status

Expiration Date

Pickup Window

Actions

Create Donation

Edit Donation

Delete Donation

View Reservation

Access

Donor Organizations

15.6 Donate Food Screen

This screen collects information about surplus food before publication.

Required Fields

Food Name

Food Category

Quantity

Measurement Unit

Expiration Date

Pickup Date

Pickup Time

Community Area

Organization

Description

Food Images

Pickup Instructions

Validation Rules

Required fields cannot be empty.

Expiration date must not be in the past.

Quantity must be greater than zero.

Images must meet supported file requirements.

Access

Donor Organizations

15.7 Available Food Screen

This is the primary screen used by recipient organizations.

Data Displayed

Food Image

Food Name

Category

Donor Organization

Community Area

Quantity

Expiration Date

Availability Status

Pickup Window

Search Fields

Food Name

Category

Organization

Community

Availability

Sorting

Newest

Closest Expiration

Recently Added

Alphabetical

Access

Recipient Organizations

15.8 Food Details Screen

Selecting a food listing displays additional information.

Data Displayed

Food Images

Description

Storage Instructions (if provided)

Pickup Address

Pickup Time

Donor Contact (limited to verified users)

Reservation Status

Actions

Reserve Food

Contact Donor

Return to Search

15.9 Reservation Screen

Displays

Donation Summary

Recipient Organization

Pickup Schedule

Reservation Status

Confirmation Checkbox

Actions

Confirm Reservation

Cancel Reservation

Access

Recipient Organizations

15.10 Pickup Schedule

Data

Pickup Date

Pickup Time

Pickup Address

Donor Organization

Recipient Organization

Status

Notes

Status Values

Scheduled

Completed

Cancelled

Access

Donor

Recipient

Administrator

15.11 Messages

Data

Conversation List

Participants

Message History

Time Sent

Read Status

Actions

Send Message

Delete Conversation (Future)

Search Messages (Future)

Access

Verified Organizations

Administrator

15.12 Notifications

Notification Types

New Donation

Reservation Received

Reservation Confirmed

Pickup Reminder

Verification Approved

System Announcement

Information Displayed

Title

Description

Time

Read Status

Actions

Mark Read

Delete

Open Related Screen

15.13 Organization Profile

Organization Information

Organization Name

Organization Type

Address

Community

Contact Person

Phone Number

Email

Operating Hours

Logo

Verification Badge

Editable Fields

Contact Information

Operating Hours

Logo

Description

Access

Organization Owner

Administrator

15.14 Administrator Dashboard

The administrator dashboard aggregates platform-wide information.

Statistics

Total Organizations

Pending Verifications

Active Donations

Completed Pickups

Communities

Daily Activity

Recent Activity

New Registrations

New Donations

Recent Reservations

Reports

Administrator Actions

Approve Organizations

Reject Organizations

View Reports

Manage Communities

Manage Users

15.15 Analytics

Metrics

Organizations Registered

Food Listings

Food Collected

Food Expired

Reservations

Monthly Activity

Community Comparison

Charts

Bar Chart

Line Chart

Pie Chart

Trend Analysis

Access

Administrator

15.16 Community Management

Community Information

Community Name

Description

Participating Organizations

Donation Statistics

Recipient Statistics

Coverage Area

Actions

Edit Community

Disable Community (Future)

Add New Community (Future)

15.17 Role-Based Data Access

Not every user should have access to the same information.

Data	Public	Donor	Recipient	Admin
------	--------	-------	-----------	-------

Home Page	✓	✓	✓	✓
Communities	✓	✓	✓	✓
Available Food	✗	✗	✓	✓
My Donations	✗	✓	✗	✓
Reservations	✗	✗	✓	✓
Analytics	✗	✗	✗	✓
Organization Management	✗	✗	✗	✓

This role-based access control ensures that users only view information relevant to their responsibilities.

15.18 Relationship Between Frontend and Backend

The frontend should not interact directly with the database. Instead, it communicates with backend services through secure APIs.

User

↓

Frontend

↓

Authentication

↓

Backend API

↓

Database

↓

Response

↓

Frontend Update

This separation improves security, maintainability, and scalability.

15.19 Why These Data Requirements?

Defining frontend data requirements before development provides several benefits:

- Ensures consistency between frontend and backend teams.
- Reduces misunderstandings during implementation.
- Clarifies what data each screen requires.
- Supports role-based access control.
- Simplifies testing by providing clear expectations.

This structured approach helps the team build a reliable and maintainable platform.

Chapter Summary

This chapter identified the data required by each major screen within the Surplus Food Redistribution Network and defined how different user roles interact with that data. It also established the relationship between the frontend and backend through secure APIs and role-based access control, providing a clear specification for implementation.

Perfect. These final chapters are what separate a **good university project** from a **professional software design document**. Many teams stop after describing the screens, but professional documentation always explains **why** certain design choices were made.

Let's continue.

CHAPTER SIXTEEN

Design Decisions and Justifications

16.1 Introduction

Every design decision made during the development of the **Surplus Food Redistribution Network** was guided by the findings of the research conducted during the analysis phase. Rather than selecting features based on assumptions, the interface, workflows, and system architecture were intentionally designed to address the operational challenges faced by organizations involved in food redistribution.

This chapter explains the reasoning behind the major design decisions and demonstrates how each decision contributes to solving the problems identified in the research.



Illustration 16.1 – Research-Driven Design

Research Findings

|



Design Decisions

|



Frontend Features

|



Better User Experience

16.2 Decision 1 — Role-Based Dashboards

Decision

Separate dashboards were designed for:

- Donor Organizations
- Recipient Organizations
- Administrators

instead of using a single dashboard for every user.

Justification

Each role performs different tasks.

For example:

Donors

- Publish food.
- Manage donations.
- Track pickups.

Recipients

- Search available food.
- Reserve donations.
- Schedule pickups.

Administrators

- Verify organizations.
- Manage communities.
- Monitor analytics.

Displaying irrelevant features would increase cognitive load and make the platform more difficult to use.

Role-based dashboards simplify navigation and improve productivity.

16.3 Decision 2 — Public Website Before Login

Decision

Visitors can browse the website before creating an account.

Justification

Organizations often want to understand:

- What the platform does.
- How it works.
- Which communities are supported.
- Whether they are eligible.

Providing public information builds trust before asking organizations to register.

16.4 Decision 3 — Verification Before Access

Decision

New organizations require administrator approval before gaining access to operational features.

Justification

The platform coordinates food redistribution between real organizations.

Verification:

- Prevents fraudulent registrations.
 - Builds trust.
 - Improves data quality.
 - Protects participating organizations.
-

16.5 Decision 4 — Interactive Community Map

Decision

The platform includes an interactive map focused on the six selected Chicago community areas.

Justification

Research indicated that location plays an important role in food redistribution.

The map helps organizations:

- Understand service coverage.
- Identify nearby partners.
- Visualize participating organizations.

The architecture also supports future expansion beyond Chicago without redesigning the platform.

16.6 Decision 5 — Responsive Design

Decision

The interface adapts to desktop, tablet, and mobile devices.

Justification

Different organizations access digital services using different devices.

For example:

- Grocery managers may use desktop computers.
- Food pantry coordinators may use tablets.
- Staff members may access the platform using smartphones while collecting food.

Responsive design improves usability across all supported devices.

16.7 Decision 6 — Component-Based Interface

Decision

The frontend is designed using reusable components.

Examples include:

- Navigation
 - Food Cards
 - Dashboard Cards
 - Search
 - Filters
 - Forms
-

Justification

Reusable components:

- Improve consistency.
- Reduce duplicated code.
- Simplify maintenance.
- Support future expansion.

This approach follows modern frontend development practices.

16.8 Decision 7 — Search and Filtering

Decision

Recipients can search and filter food listings.

Justification

Food donations are time-sensitive.

Organizations should not need to scroll through every listing to find suitable food.

Filters improve efficiency by allowing users to search by:

- Category
- Community

- Organization
 - Availability
-

16.9 Decision 8 — Notifications

Decision

Real-time notifications inform users about important events.

Justification

Research highlighted communication delays as a significant challenge.

Notifications reduce missed opportunities by informing users when:

- New donations become available.
 - Reservations are confirmed.
 - Pickup times change.
 - Accounts are approved.
-

16.10 Decision 9 — Simple Navigation

Decision

Navigation was intentionally designed to be straightforward and consistent.

Justification

Many participating organizations may have limited technical experience.

Simple navigation reduces confusion and shortens the learning curve.

16.11 Decision 10 — Accessibility

Decision

Accessibility considerations were incorporated throughout the interface.

Justification

The platform should be usable by the widest possible range of users.

Accessibility features include:

- Readable typography.
 - Keyboard navigation.
 - High-contrast colors.
 - Accessible forms.
 - Responsive layouts.
-

16.12 Decision 11 — Scalable Architecture

Decision

Although the MVP focuses on six Chicago community areas, the architecture supports future expansion.

Justification

The BUILD project specifies Chicago as the initial deployment area.

However, the platform is designed so that additional:

- Communities
- Cities
- States
- Countries

can be added without redesigning the frontend.

16.13 Decision 12 — Free and Open Technologies

Decision

The project prioritizes free and widely available technologies.

Examples include:

- HTML
 - CSS
 - JavaScript
 - Firebase (Spark plan for the prototype)
 - Leaflet with OpenStreetMap for mapping
-

Justification

Using free technologies:

- Reduces development costs.
 - Makes collaboration easier.
 - Allows the project to be reproduced by other teams.
 - Supports continued development after the BUILD program.
-

16.14 Alignment with Research

The table below demonstrates how the design decisions directly respond to research findings.

Research Finding	Design Decision
Food expires before redistribution	Fast donation workflow
Organizations struggle to discover food	Search, filters, interactive map
Communication delays	Notifications and messaging
Trust is essential	Organization verification
Different users perform different tasks	Role-based dashboards
Platform may expand in future	Scalable architecture

This alignment ensures that the solution remains evidence-based rather than feature-driven.

16.15 Why These Decisions Matter

Collectively, these design decisions contribute to a platform that is:

- Easy to learn.
- Efficient to use.
- Accessible.
- Scalable.
- Secure.
- Maintainable.

Each decision supports the overall objective of reducing food waste while improving coordination between participating organizations.

Chapter Summary

This chapter explained the rationale behind the major design decisions made during the development of the Surplus Food Redistribution Network. Each decision was linked to challenges identified in the research and justified based on usability, accessibility, scalability, and maintainability considerations. The result is a design that is purposeful, evidence-based, and aligned with the goals of the BUILD project.

CHAPTER SEVENTEEN

Future Enhancements

17.1 Introduction

The current implementation of the **Surplus Food Redistribution Network** focuses on delivering a practical Minimum Viable Product (MVP) that addresses the challenges identified during the research phase within the six selected Chicago community areas.

However, food redistribution is a complex and evolving process. As additional organizations join the platform and operational requirements grow, new features can be introduced to improve efficiency, increase participation, and expand the platform's reach.

The enhancements described in this chapter are not part of the current BUILD project implementation but represent potential future developments.



Illustration 17.1 – Platform Growth

MVP

|



Additional Features

|



Regional Expansion

|



National Platform

|



Global Platform

17.2 Multi-City Expansion

Current MVP

The platform currently supports only the six selected community areas within Chicago.

Future Enhancement

The system can be expanded to support:

- Additional Chicago communities
- Other cities in Illinois
- Other U.S. states
- International deployments

Benefits

- Larger network of organizations.
- Greater food redistribution impact.
- Increased scalability.
- Broader community participation.

The database and frontend architecture were intentionally designed so that geographic information can be expanded without redesigning the platform.

17.3 Volunteer Delivery System

Current MVP

Food collection is coordinated directly between donor and recipient organizations.

Future Enhancement

Introduce verified volunteer drivers who can assist with transportation.

Volunteer features may include:

- Accept delivery requests.
 - View pickup locations.
 - Navigate optimized routes.
 - Update delivery status.
 - Confirm completed deliveries.
-

Benefits

- Helps organizations without transportation.
- Reduces missed pickups.
- Expands participation.

Illustration 17.2 – Volunteer Workflow

Donor

↓

Volunteer Accepts

↓

Pickup

↓

Delivery

↓

Recipient

17.4 AI Recommendation System

Future versions could use Artificial Intelligence to improve food matching.

Possible recommendations include:

- Suggest recipient organizations based on previous donations.
 - Recommend nearby organizations.
 - Predict which donations require urgent attention.
 - Identify organizations with similar donation patterns.
-

Benefits

- Faster matching.
- Reduced food waste.
- Improved efficiency.
- Better user experience.

17.5 Predictive Analytics

Historical platform data could be analyzed to predict future demand.

Examples include:

- Seasonal food demand.
- Frequently requested food categories.
- Community-specific shortages.
- Peak donation periods.

Administrators could use these insights to make better operational decisions.

17.6 Route Optimization

As participation increases, delivery logistics become more important.

Future versions could include route optimization for pickups and deliveries.

Potential capabilities:

- Shortest route calculation.
- Traffic-aware routing.
- Multiple pickup scheduling.
- Estimated arrival times.

This enhancement would reduce transportation time and operational costs.

17.7 SMS and Email Notifications

The MVP focuses on in-platform notifications.

Future versions could additionally support:

- SMS reminders.

- Email confirmations.
- Pickup alerts.
- Reservation confirmations.
- Account verification updates.

Providing notifications through multiple communication channels would improve reliability and user engagement.

17.8 Mobile Application

Although the current platform is responsive and accessible through mobile browsers, a dedicated mobile application could provide an improved experience.

Potential platforms include:

- Android
- iOS

Additional mobile capabilities could include:

- Push notifications.
 - Offline access to selected information.
 - Camera integration for food image uploads.
 - GPS location services.
-

17.9 Multilingual Support

Future deployments may involve organizations serving diverse communities.

Potential language support could include:

- English
- Spanish
- Other languages based on community needs

Providing multilingual interfaces would improve accessibility and encourage wider participation.

17.10 Advanced Analytics Dashboard

The current administrator dashboard provides operational statistics.

Future versions may include:

- Community comparison dashboards.
- Food category trends.
- Monthly growth reports.
- Donation forecasting.
- Organization performance metrics.
- Interactive reporting.

These insights would support long-term planning and resource allocation.

17.11 Public API

A secure API could allow approved partner organizations and third-party systems to integrate with the platform.

Potential integrations include:

- Food bank management systems.
- Municipal services.
- Nonprofit organizations.
- Community resource platforms.

All integrations should follow appropriate authentication and authorization controls.

17.12 Document Management

Future versions could support document uploads for organization verification.

Examples include:

- Registration certificates.
- Operating licenses.
- Tax-exempt documentation.

- Health and safety certifications.

This would simplify the verification process and improve administrative efficiency.

17.13 Enhanced Organization Profiles

Organizations could maintain richer profiles that include:

- Operating hours.
- Accepted food categories.
- Storage capabilities.
- Refrigeration availability.
- Delivery capacity.
- Service areas.

This additional information would improve matching between donors and recipients.

17.14 Sustainability Dashboard

Future versions could calculate and display environmental impact metrics.

Examples include:

- Estimated food waste prevented.
- Estimated meals redistributed.
- Carbon emissions reduced.
- Community impact over time.

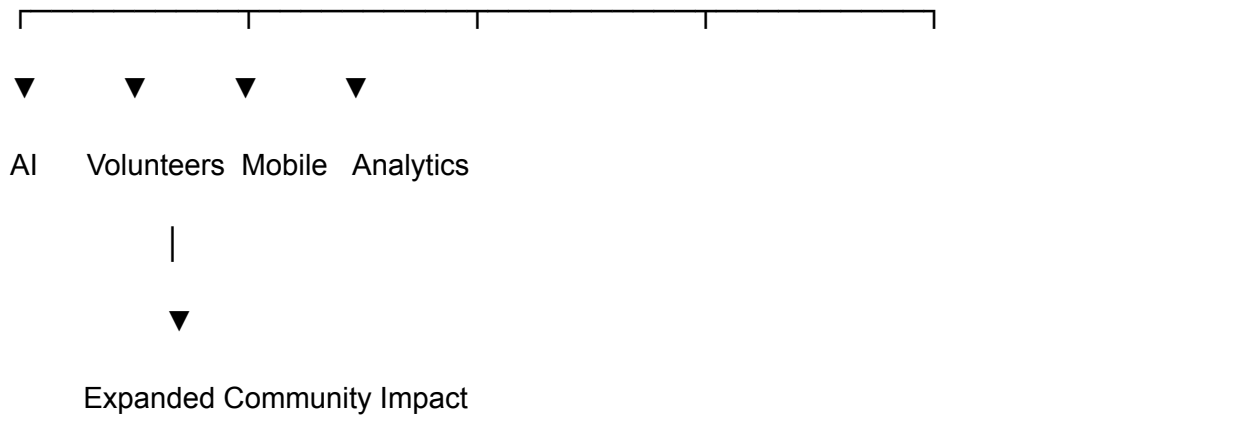
Such information would help demonstrate the social and environmental value of the platform.



Illustration 17.3 – Future Ecosystem

Future Platform

|



17.15 Prioritization Roadmap

To guide future development, enhancements can be grouped into phases.

Phase	Enhancements
Phase 1 (Current MVP)	Registration, Verification, Food Listings, Reservations, Pickups, Messaging, Dashboard
Phase 2	Volunteer Delivery, SMS/Email Notifications, Enhanced Analytics
Phase 3	AI Recommendations, Predictive Analytics, Route Optimization
Phase 4	Multi-City Expansion, Multilingual Support, Public API, Mobile Applications

This phased roadmap helps ensure that development remains focused while providing a clear path for future growth.

17.16 Why These Enhancements?

The proposed enhancements were selected because they build naturally on the existing platform without requiring a complete redesign.

The current architecture supports future expansion by:

- Using modular frontend components.
- Separating frontend and backend responsibilities.
- Applying role-based access control.
- Designing scalable navigation and information architecture.

This ensures that future features can be introduced incrementally while maintaining a consistent user experience.

Chapter Summary

This chapter presented potential future enhancements for the Surplus Food Redistribution Network beyond the scope of the current MVP. These enhancements include geographic expansion, volunteer delivery support, AI-powered recommendations, predictive analytics, route optimization, multilingual interfaces, mobile applications, and richer reporting capabilities. Together, they provide a strategic roadmap for evolving the platform while preserving the scalable foundation established in the current design.

Perfect. This will be the closing chapter of your document. A strong conclusion should not introduce new features. Instead, it should summarize the work, reinforce how the design addresses the research findings, and end on a forward-looking note.

CHAPTER EIGHTEEN

Conclusion

18.1 Introduction

The **Surplus Food Redistribution Network** was designed to address one of the most significant challenges identified during the research phase: the disconnect between organizations with surplus food and organizations serving individuals experiencing food insecurity.

Through a user-centered design process, the platform transforms research findings into a practical digital solution that enables verified organizations to coordinate food donations more efficiently, reduce unnecessary food waste, and improve access to surplus food within local communities.

Rather than functioning as a simple listing platform, the proposed solution provides a structured ecosystem where donor organizations, recipient organizations, and administrators can collaborate through a secure, accessible, and scalable interface.

Illustration 18.1 – Project Journey

Research



Problem Analysis



User Personas



Information Architecture



User Flows



Interface Design



Functional Platform



Community Impact

18.2 Research-Driven Design

A central objective of this project was to ensure that every design decision was supported by evidence gathered during the research phase.

The research highlighted several key challenges:

- Food is frequently discarded before it can be redistributed.
- Organizations lack visibility into available surplus food.
- Communication between donors and recipients can be slow and fragmented.
- Trust and verification are essential when coordinating food redistribution.
- Existing processes often rely on manual coordination, making them inefficient.

Each of these findings directly influenced the design of the platform. Features such as role-based dashboards, organization verification, searchable food listings, notifications, and structured pickup workflows were included because they respond to these identified needs rather than introducing unnecessary complexity.

18.3 Achieving the Project Objectives

The proposed frontend design meets the primary objectives established at the beginning of the project.

These objectives include:

- Providing an intuitive interface for donor and recipient organizations.
- Reducing barriers to food donation and reservation.
- Improving coordination through structured workflows.
- Supporting secure access through organization verification.
- Delivering a responsive and accessible experience across multiple devices.
- Creating a scalable foundation for future growth.

Together, these objectives contribute to a platform that is practical, efficient, and aligned with the operational needs identified in the research.

18.4 Alignment with the BUILD Project

The design aligns closely with the goals of the BUILD Capstone Project by emphasizing collaboration, practical problem-solving, and real-world applicability.

Rather than focusing solely on technical implementation, the project demonstrates how research, user experience design, and software engineering can be integrated to create a meaningful solution to a community challenge.

The documentation developed throughout this project provides a clear blueprint that can guide both frontend and backend development while maintaining consistency across the team.

18.5 Contribution to Sustainable Development Goal 2

The platform directly supports the objectives of **United Nations Sustainable Development Goal 2 (Zero Hunger)** by improving access to surplus food that might otherwise be wasted.

Although the project is implemented as a prototype focused on six selected community areas in Chicago, the underlying architecture has been designed to support future expansion into additional communities and regions.

By improving coordination between food donors and recipient organizations, the platform has the potential to:

- Reduce avoidable food waste.
 - Increase access to surplus food.
 - Strengthen collaboration between community organizations.
 - Improve the efficiency of food redistribution efforts.
-

18.6 Strengths of the Proposed Design

Several characteristics contribute to the strength of the proposed solution.

The platform is:

- User-centered, with decisions based on research findings.
- Role-based, providing tailored experiences for different users.
- Responsive, supporting desktop, tablet, and mobile devices.
- Accessible, following inclusive design principles.
- Modular, enabling easier maintenance and future development.
- Scalable, supporting expansion beyond the initial deployment area.

These qualities provide a strong foundation for future development while maintaining a focus on usability and maintainability.

18.7 Project Limitations

As with any Minimum Viable Product (MVP), the current implementation has defined boundaries.

The project focuses on:

- Six selected Chicago community areas.
- Verified organizations rather than individual users.
- Basic food redistribution workflows.
- Core administrative functionality.

Features such as volunteer delivery management, predictive analytics, multilingual support, mobile applications, and AI-assisted recommendations have been identified as future enhancements rather than part of the initial implementation.

Clearly defining the scope ensures that the project remains achievable while providing opportunities for continued development.

18.8 Reflection on the Design Process

The design process demonstrated the importance of beginning with research before creating interfaces.

Rather than designing screens first and identifying problems later, the project followed a structured workflow:

1. Understand the problem.
2. Identify the users.
3. Analyze user needs.
4. Define system requirements.
5. Design information architecture.
6. Create user flows.
7. Develop interface designs.
8. Plan implementation.

This approach resulted in a solution where every major feature can be traced back to a documented user need or operational challenge.

Illustration 18.2 – Human-Centered Design Process

Understand Users

↓

Research

↓

Define Problems

↓

Design Solution

↓

Prototype

↓

Build

↓

Evaluate

↓

Improve

18.9 Final Remarks

The **Surplus Food Redistribution Network** demonstrates how thoughtful user experience design and research-driven decision-making can contribute to addressing real-world social challenges.

By connecting donor organizations with recipient organizations through a secure, intuitive, and scalable platform, the proposed solution aims to simplify food redistribution while reducing unnecessary waste and strengthening community collaboration.

Although developed as part of the BUILD Capstone Project, the design principles, workflows, and architectural decisions described in this document provide a foundation that could support future real-world implementation with additional development and stakeholder collaboration.

Ultimately, the success of a platform is measured not only by its technical capabilities but also by its ability to solve meaningful problems for the people who use it. The proposed design reflects this philosophy by placing users, accessibility, and community impact at the center of every design decision.